Formal Grammar

Concepts of Programming Languages

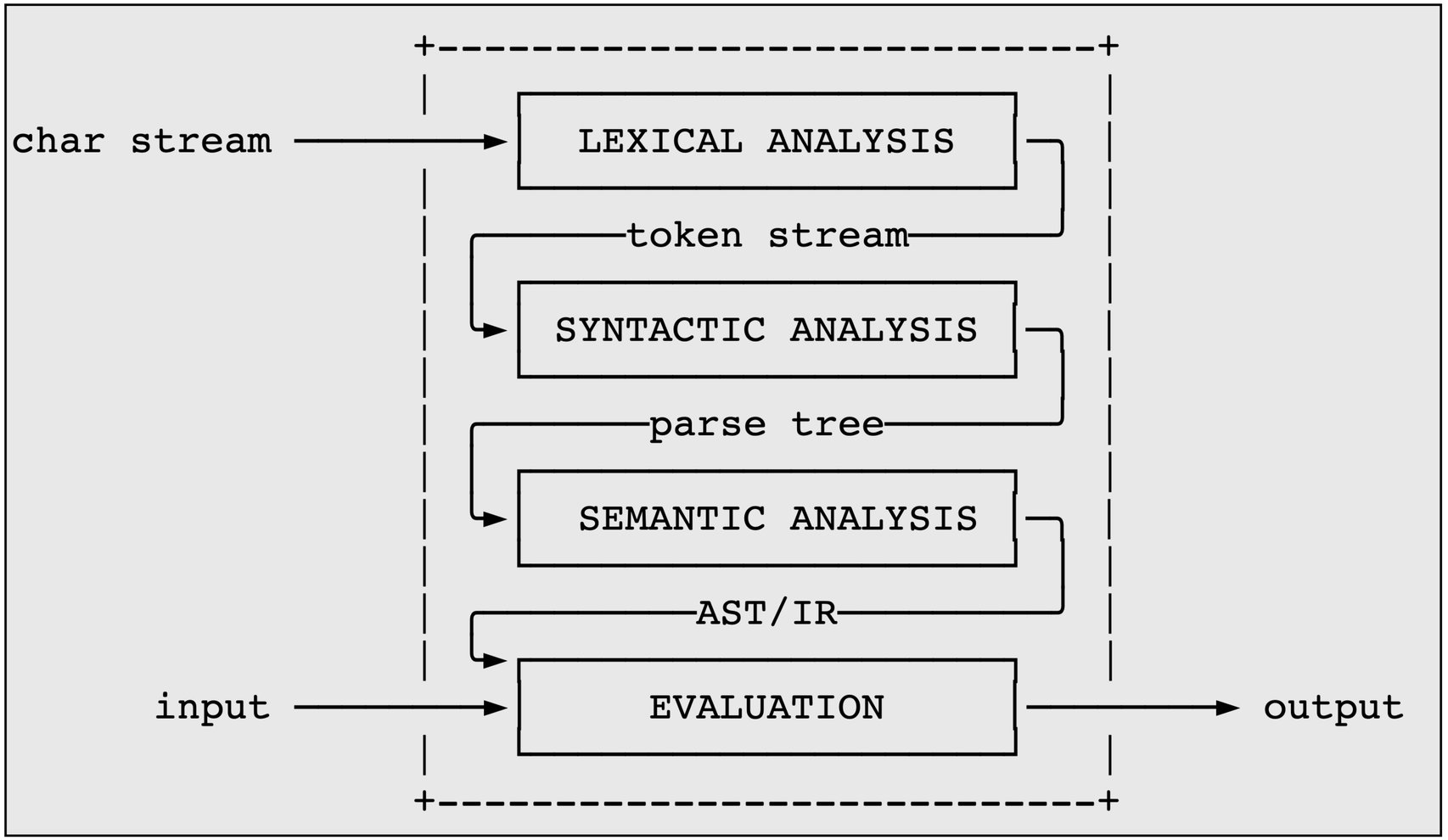
CAS CS 320

Outline

- Discuss briefly the **interpretation pipeline**
- Introduce **formal grammars,** a mathematical framework for syntax and parsing
- Discuss grammatical **ambiguity**

The Interpretation Pipeline

The Picture


char stream
LEXICAL ANALYSIS
token stream
SYNTACTIC ANALYSIS
parse tree
SEMANTIC ANALYSIS
AST/IR
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EVALUATION
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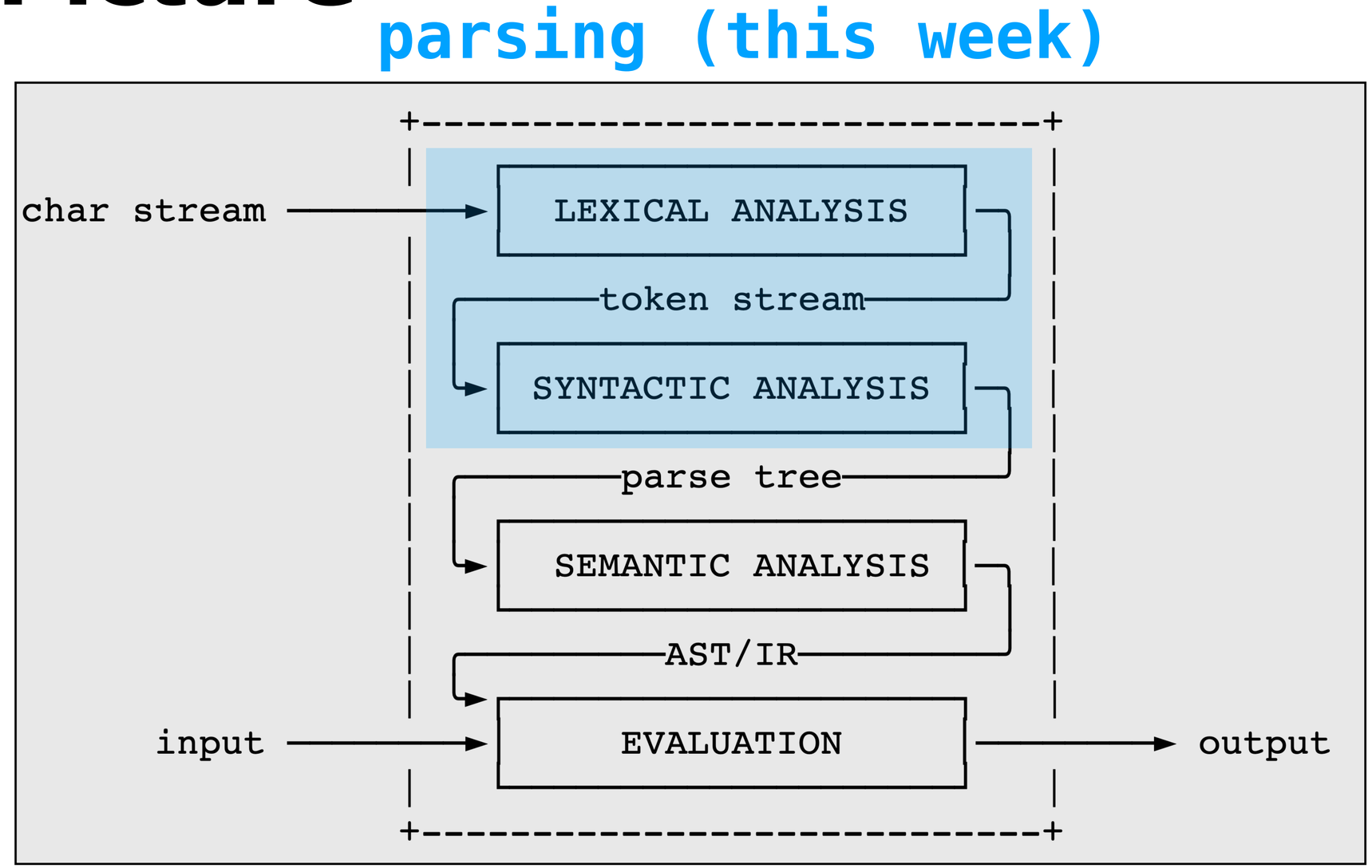
The Picture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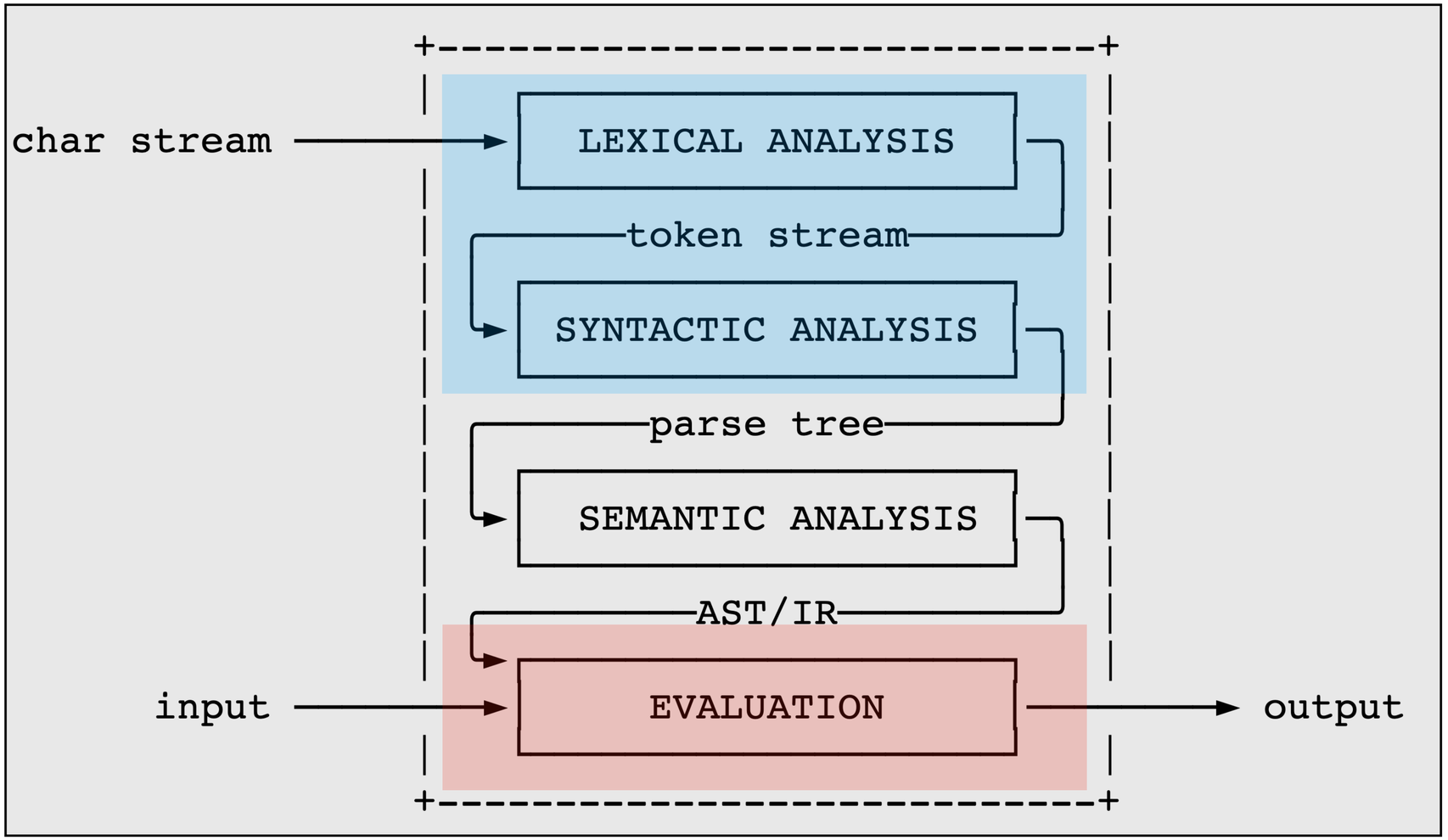

semantics (after the break)

A Note on Compilation

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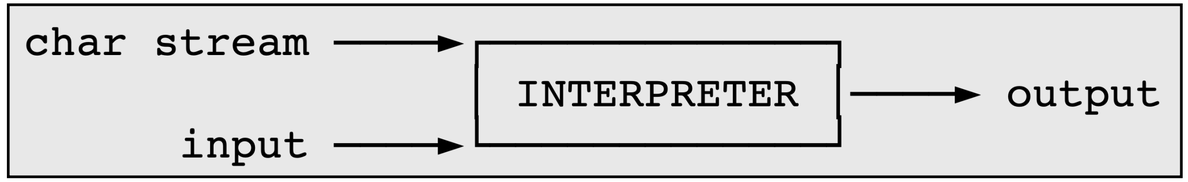
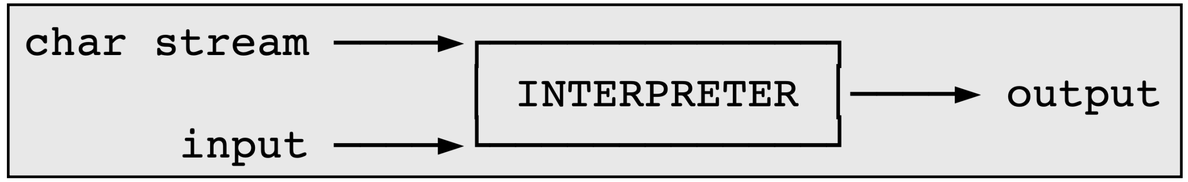
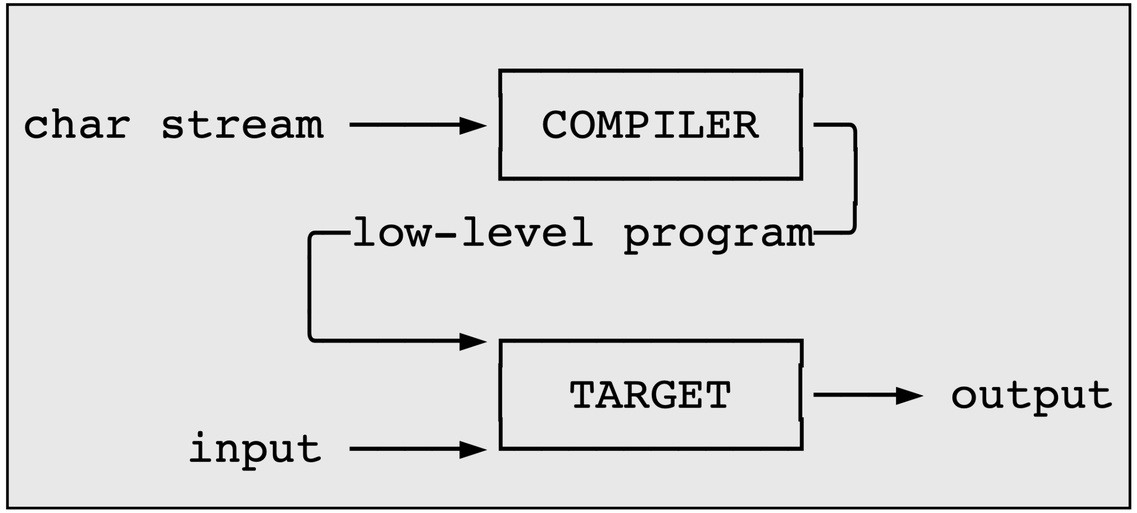


We'll be building programs that read and evaluate programs (**interpreters**)

A Note on Compilation





We'll be building programs that read and evaluate programs (**interpreters**)

In a different course you might *translate* programs into another language (**compilers,** we'll cover this briefly)

The Mini-Projects

There will be **three** mini-projects, each 2 weeks long

For each project, you'll build an interpreter (calculator) and you'll be given:

- » syntax rules
- » typing rules
- » semantic rules

Today

We need a formal language for describing the syntax of programming languages

This is part of the study of **formal language theory**

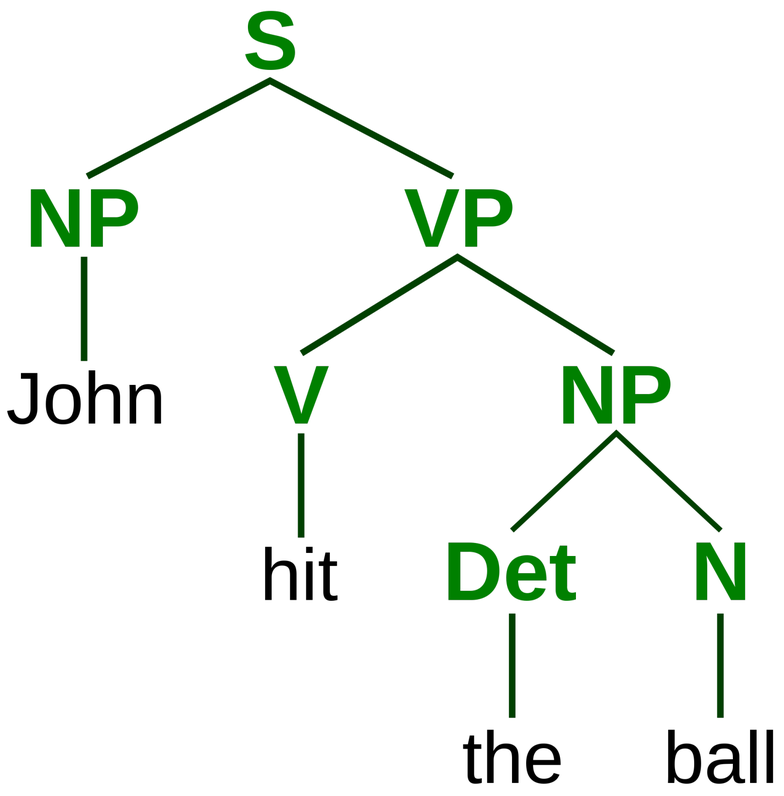
Nearly every PL out there (including OCaml) is described using **Backus–Naur Form (BNF) Grammars**

Formal Grammar

What is Grammar?

John hit the ball

⇓



https://commons.wikimedia.org/wiki/File:ParseTree.svg

What is Grammar?

Grammar refers to the rules which govern what statements are well-formed

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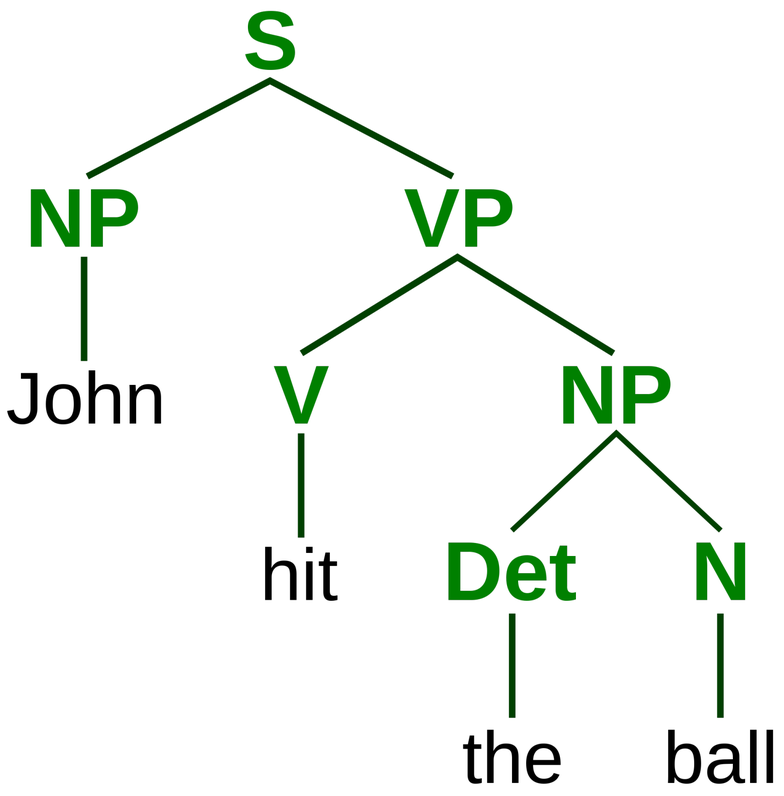

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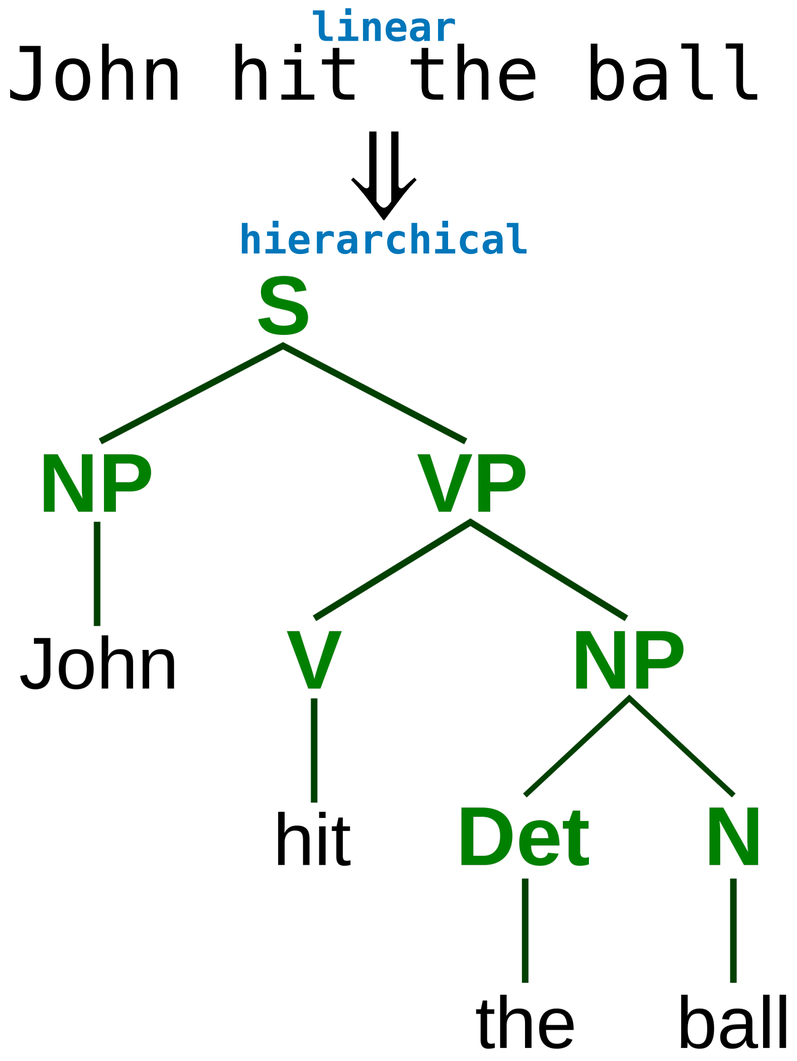
What is Grammar?

Grammar refers to the rules which govern what statements are well-formed

Grammar gives **linear** statements (in natural language or code) their **hierarchical** structure



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Grammar vs. Semantics

I taught my car in the refrigerator. ✓

vs.

My the car taught I refrigerator. ✗

Grammar is not about meaning, just structure

(As we will see, it is useful to separate these two concerns)

Grammars for Programming Languages

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Formal grammars for PL
tell us which programs
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Grammars for Programming Languages

```
# let f x = x + 1;;
val f : int -> int = <fun>
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Grammars for Programming Languages

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val f : int -> int = <fun>
# let rec x = x x x x ;;
Line 1, characters 14-15:
1 | let rec x = x x x x ;;
                  ^
Error: This expression has type ...
       but an expression was ex ...
       The type variable 'a occ ...
```

Grammars for Programming Languages

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Line 1, characters 14-15:
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Error: This expression has type ...
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# let rec f x = f x + 1 - 1;;
val f : 'a -> int = <fun>
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Grammars for Programming Languages

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val f : 'a -> int = <fun>
# let x = List.hd [];;
Exception: Failure "hd".
```

Grammars for Programming Languages

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(In OCaml, well-formed programs are the ones we can type-check)

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# let rec x = x x x x ;;
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Error: This expression has type ...
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# let rec f x = f x + 1 - 1;;
val f : 'a -> int = <fun>
# let x = List.hd [];;
Exception: Failure "hd".
# let x = ;;
Line 1, characters 8-10:
1 | let x = ;;
            ^^
Error: Syntax error
```

How do we formally represent well-formed sentences?

An Example

```
the cow jumped over the moon
```

An Example

the cow jumped over the moon

How do we know this a well-formed sentence?

An Example

```
<article> cow jumped over the moon
```

An Example

<article> <noun> jumped over the moon

An Example

<noun-phrase> jumped over the moon

An Example

<noun-phrase> jumped over <article> moon

An Example

<noun-phrase> jumped over <article> <noun>

An Example

<noun-phrase> jumped over <noun-phrase>

An Example

`<noun-phrase> jumped over <noun-phrase>`

a thing jumped over a thing

An Example

<noun-phrase> jumped <prep> <noun-phrase>

An Example

<noun-phrase> jumped <prep-phrase>

An Example

<noun-phrase> <verb> <prep-phrase>

An Example

<noun-phrase> <verb-phrase>

An Example

<noun-phrase> <verb-phrase>

a thing did a thing

An Example

<sentence>

An Example

<sentence>

We know it's a sentence because it has the right kind of hierarchical structure

A Derivation

```
<sentence>
<noun-phrase> <verb-phrase>
<noun-phrase> <verb> <prep-phrase>
<noun-phrase> jumped <prep-phrase>
<noun-phrase> jumped <prep> <noun-phrase>
<noun-phrase> jumped over <noun-phrase>
<noun-phrase> jumped over <article> <noun>
<noun-phrase> jumped over <article> moon
<noun-phrase> jumped over the moon
<article> <noun> jumped over the moon
<article> cow jumped over the moon
the cow jumped over the moon
```

A Derivation

```
<sentence>
<noun-phrase> <verb-phrase>
<noun-phrase> <verb> <prep-phrase>
<noun-phrase> jumped <prep-phrase>
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<noun-phrase> jumped over <noun-phrase>
<noun-phrase> jumped over <article> <noun>
<noun-phrase> jumped over <article> moon
<noun-phrase> jumped over the moon
<article> <noun> jumped over the moon
<article> cow jumped over the moon
```

```
the cow jumped over the moon
```

A Derivation

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<noun-phrase> jumped over <article> <noun>
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<noun-phrase> jumped over the moon
<article> <noun> jumped over the moon
<article> cow jumped over the moon
```

the cow jumped over the moon

A Derivation

<sentence>
<noun-phrase> <verb-phrase>
<noun-phrase> <verb> <prep-phrase>
<noun-phrase> jumped <prep-phrase>
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<noun-phrase> jumped over <noun-phrase>
<noun-phrase> jumped over <article> <noun>
<noun-phrase> jumped over <article> moon
<noun-phrase> jumped over the moon
<article> <noun> jumped over the moon

<article>

the cow jumped over the moon

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```
<article>
```

the cow jumped over the moon

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```
<article> <noun>

the cow jumped over the moon
```

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<noun-phrase>
<article> <noun>
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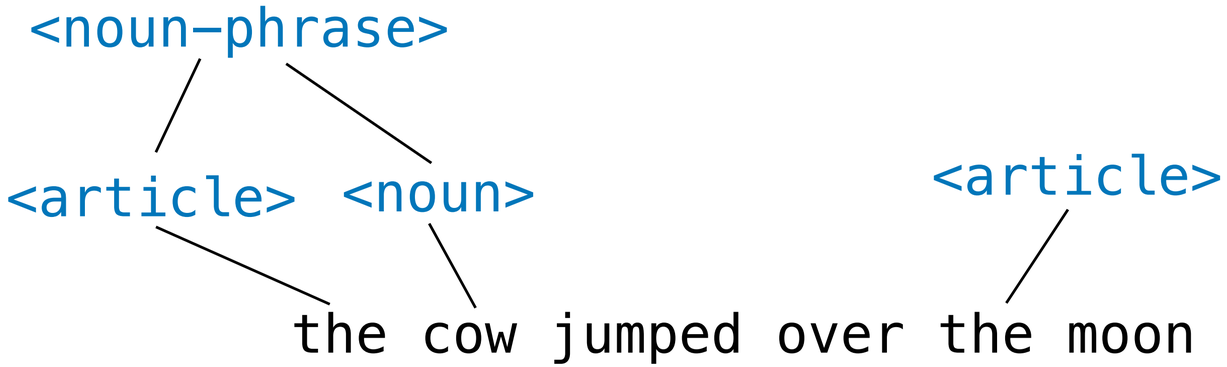

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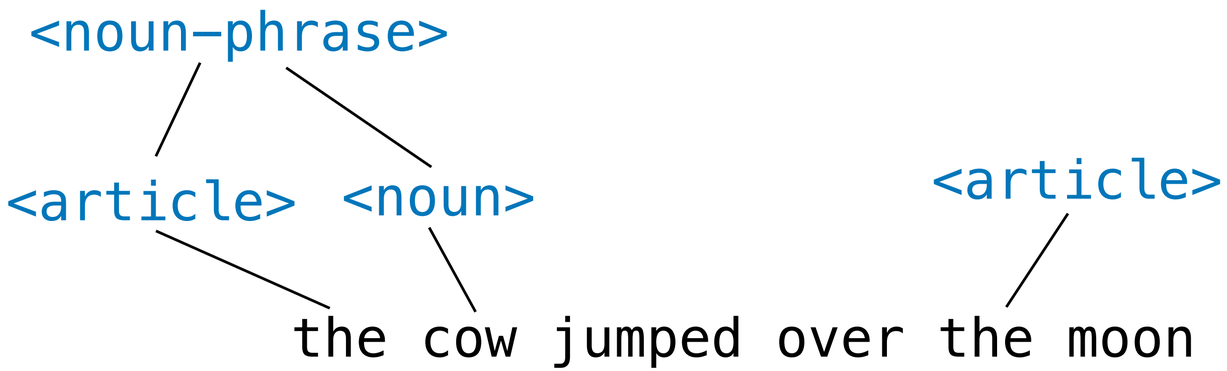

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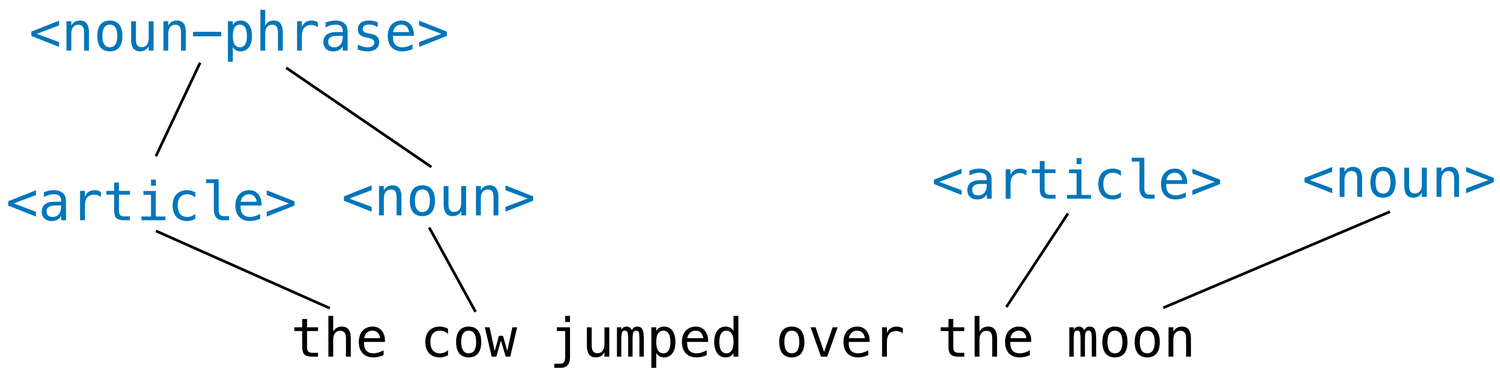

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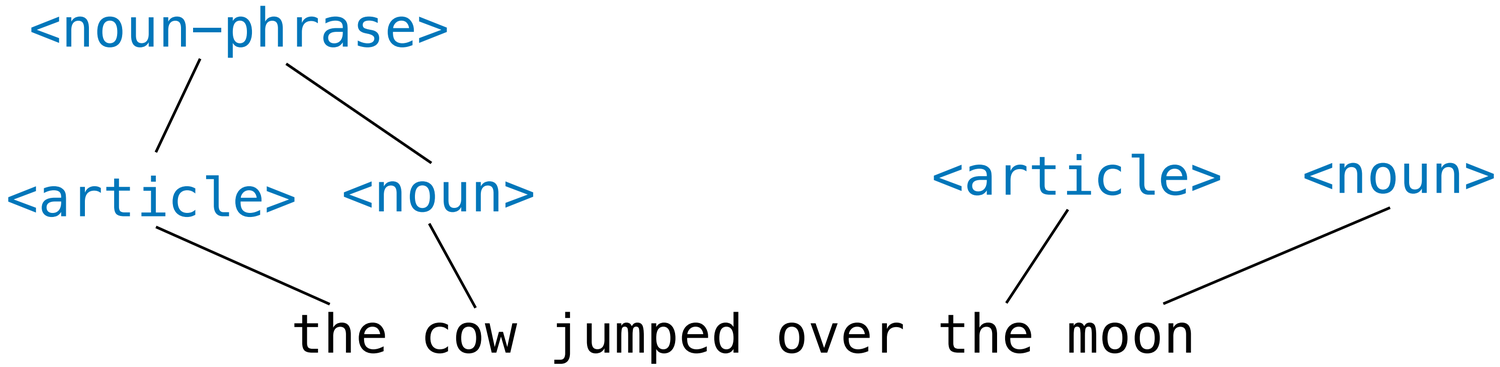

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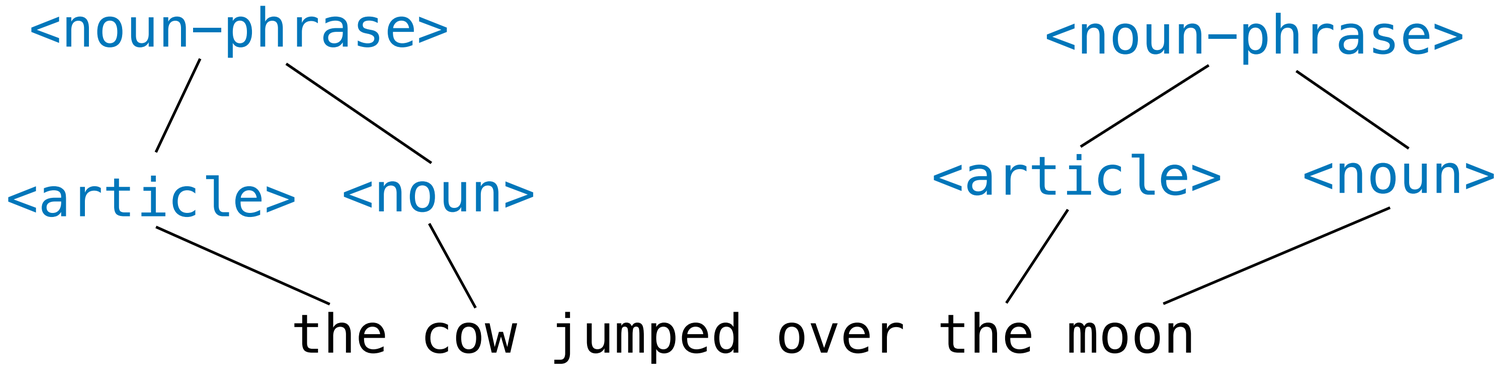

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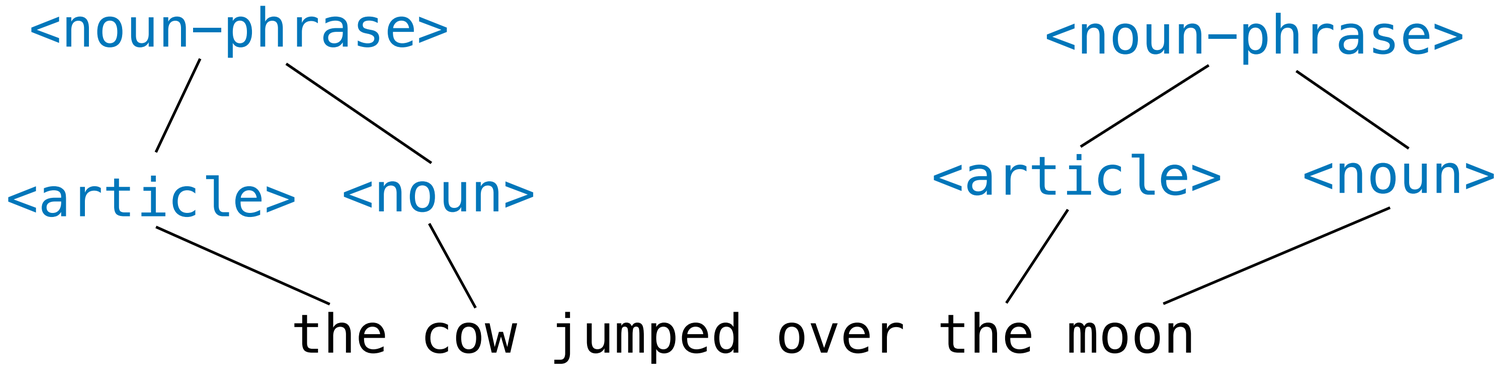

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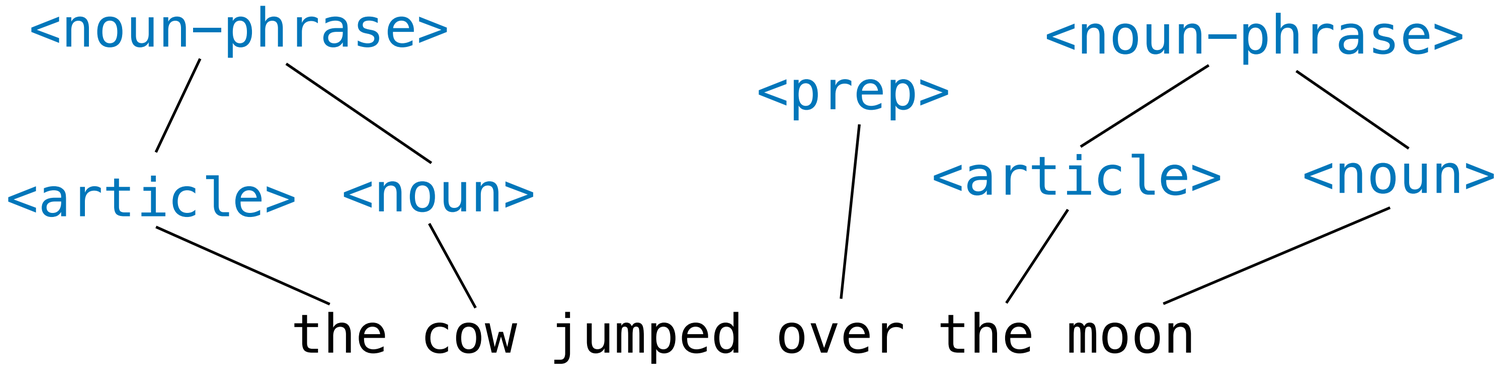

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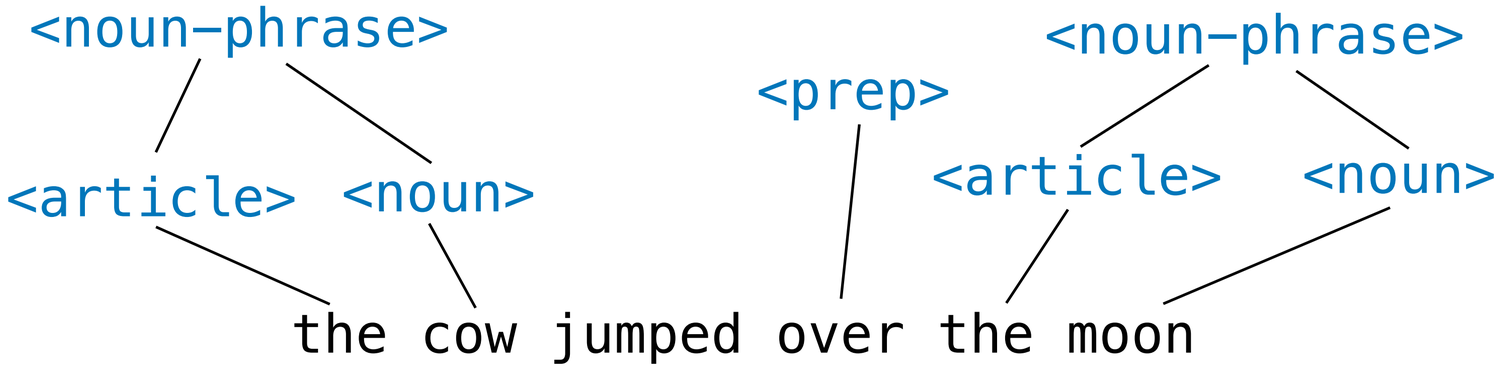

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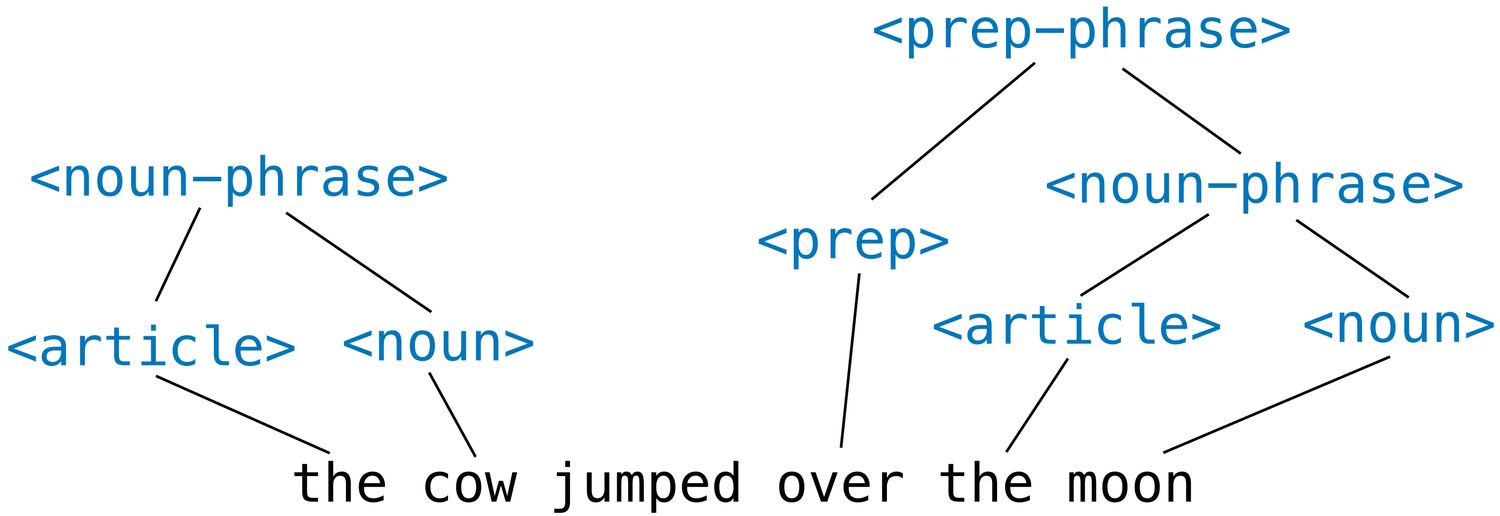

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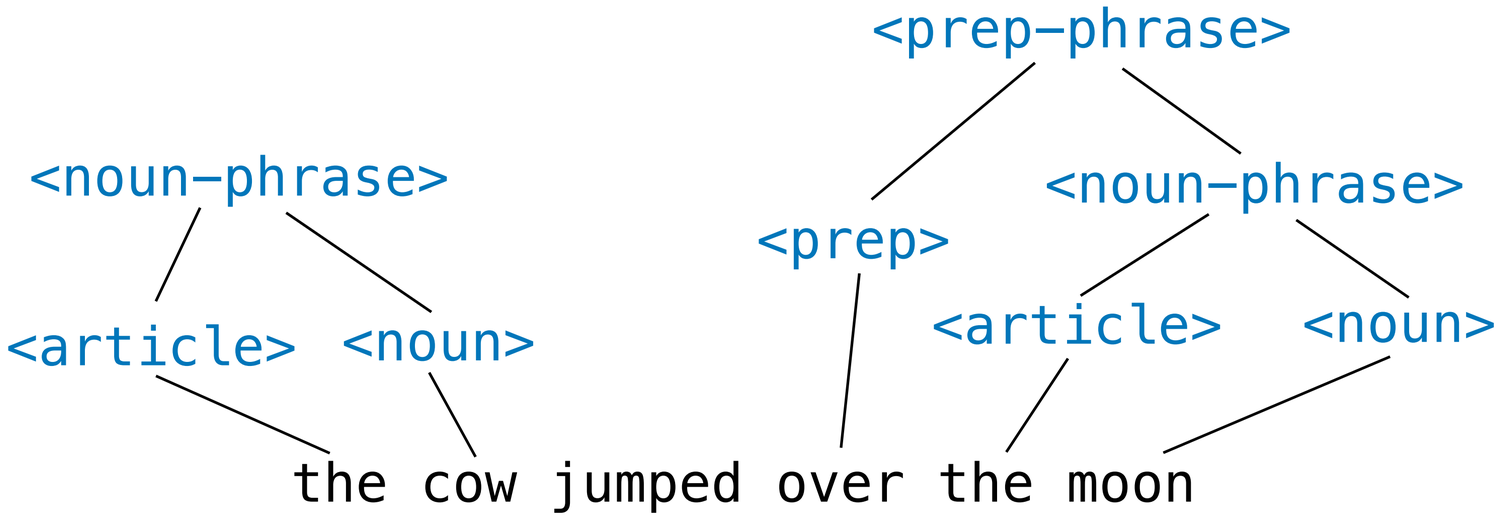

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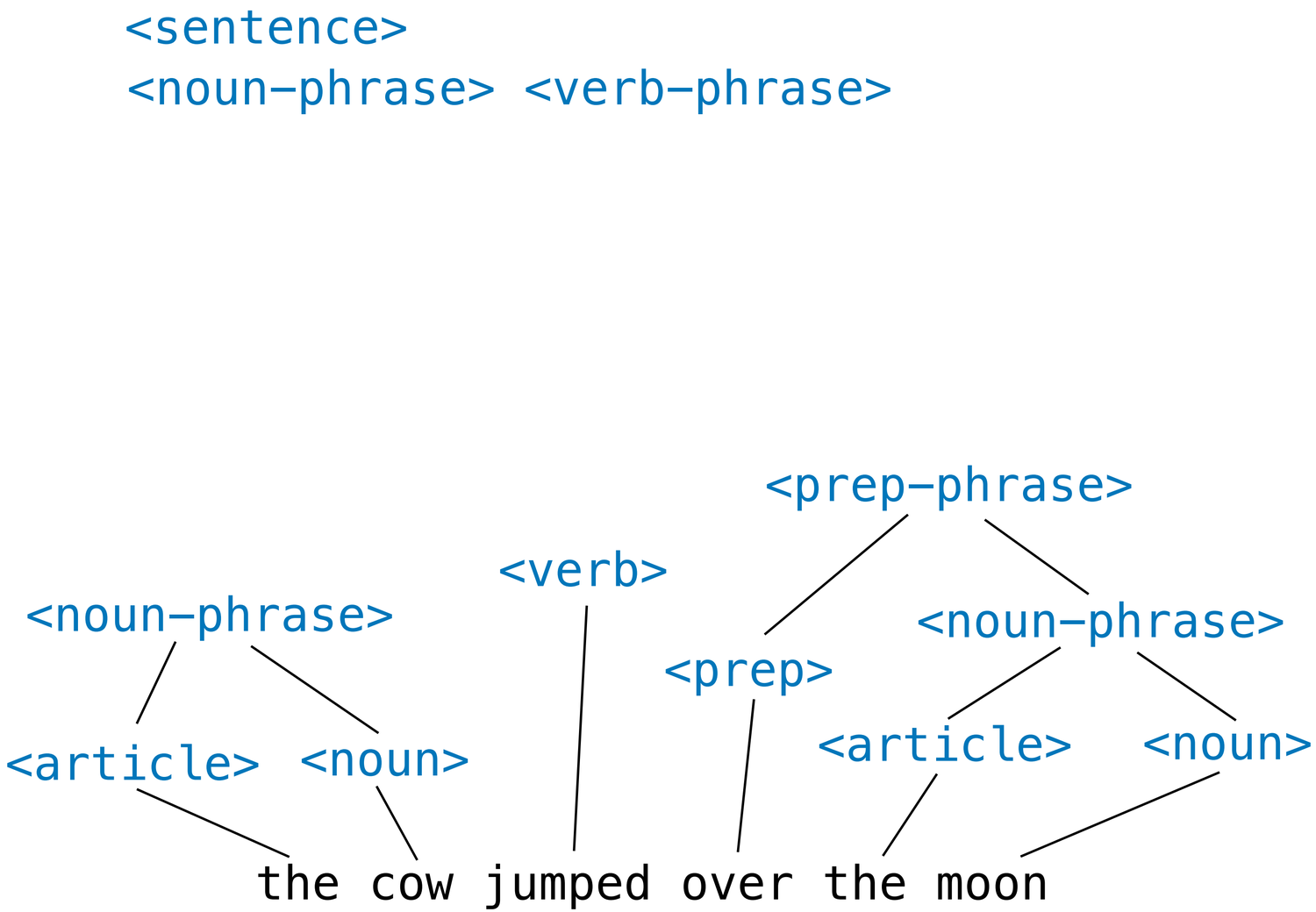

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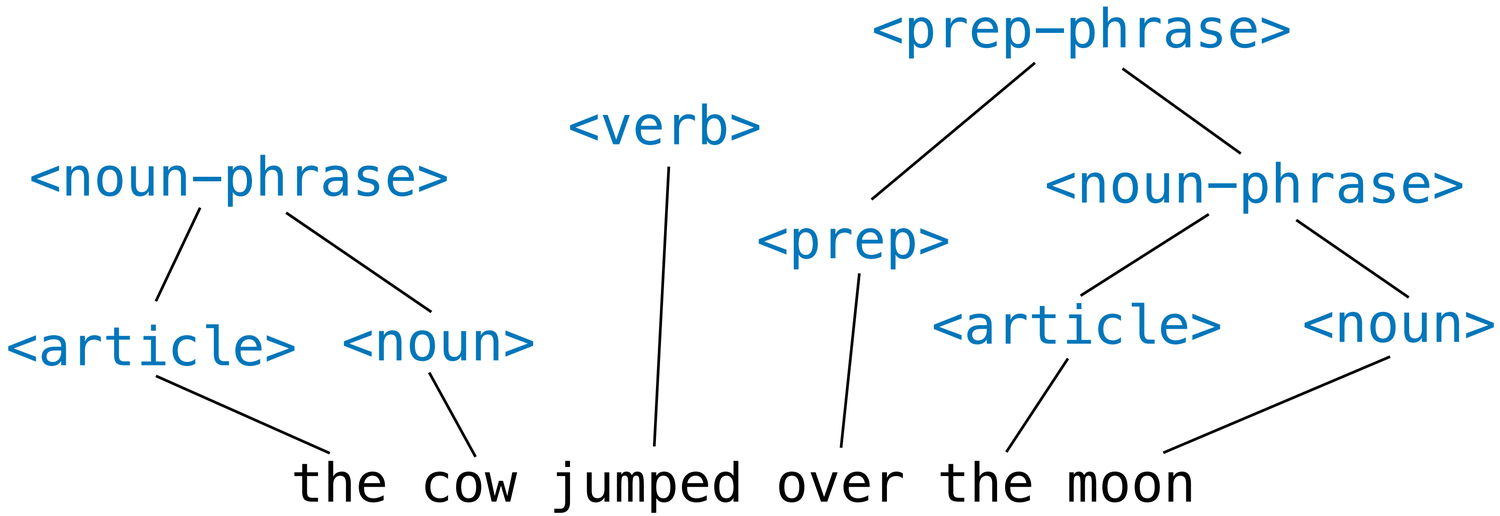

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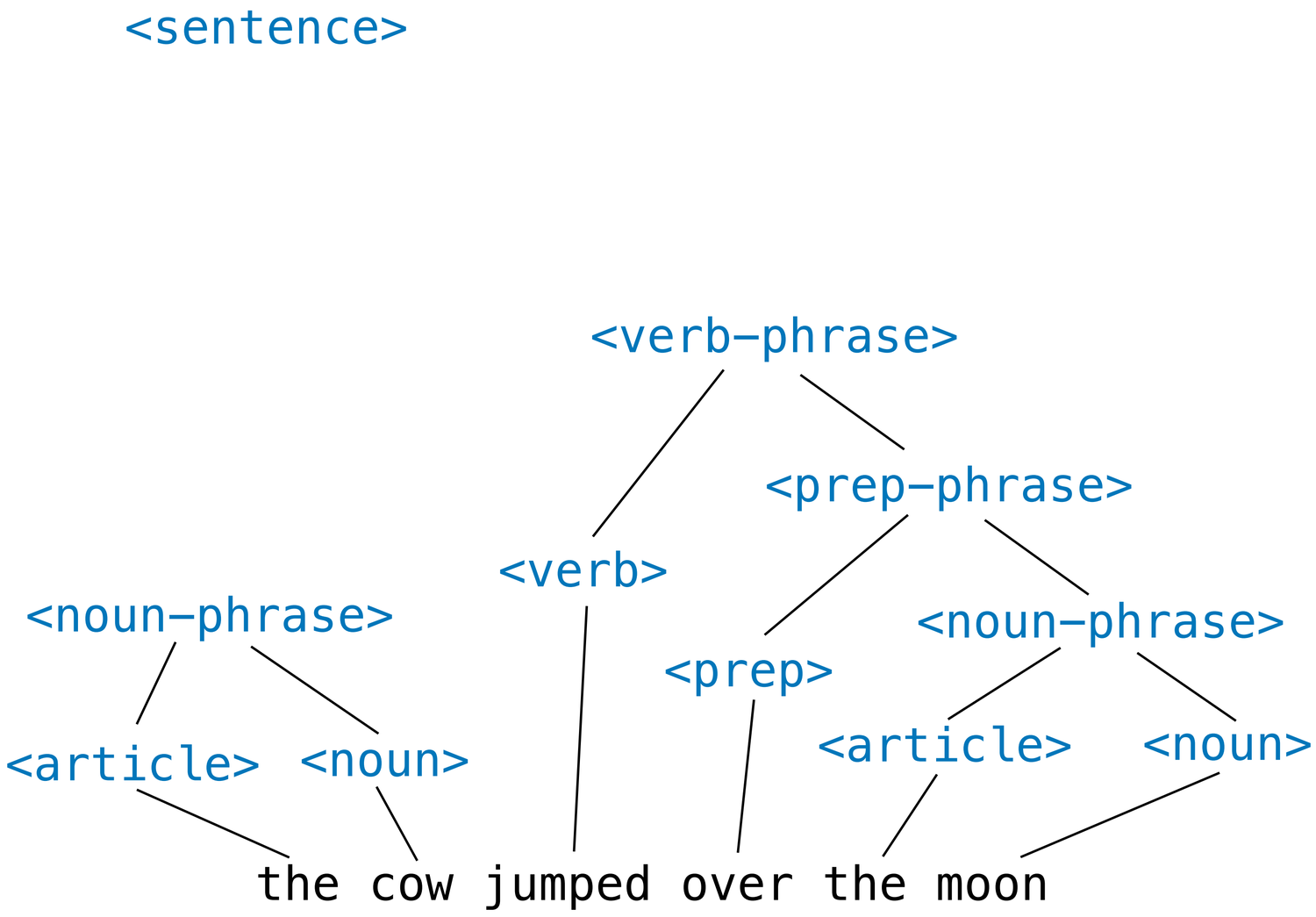

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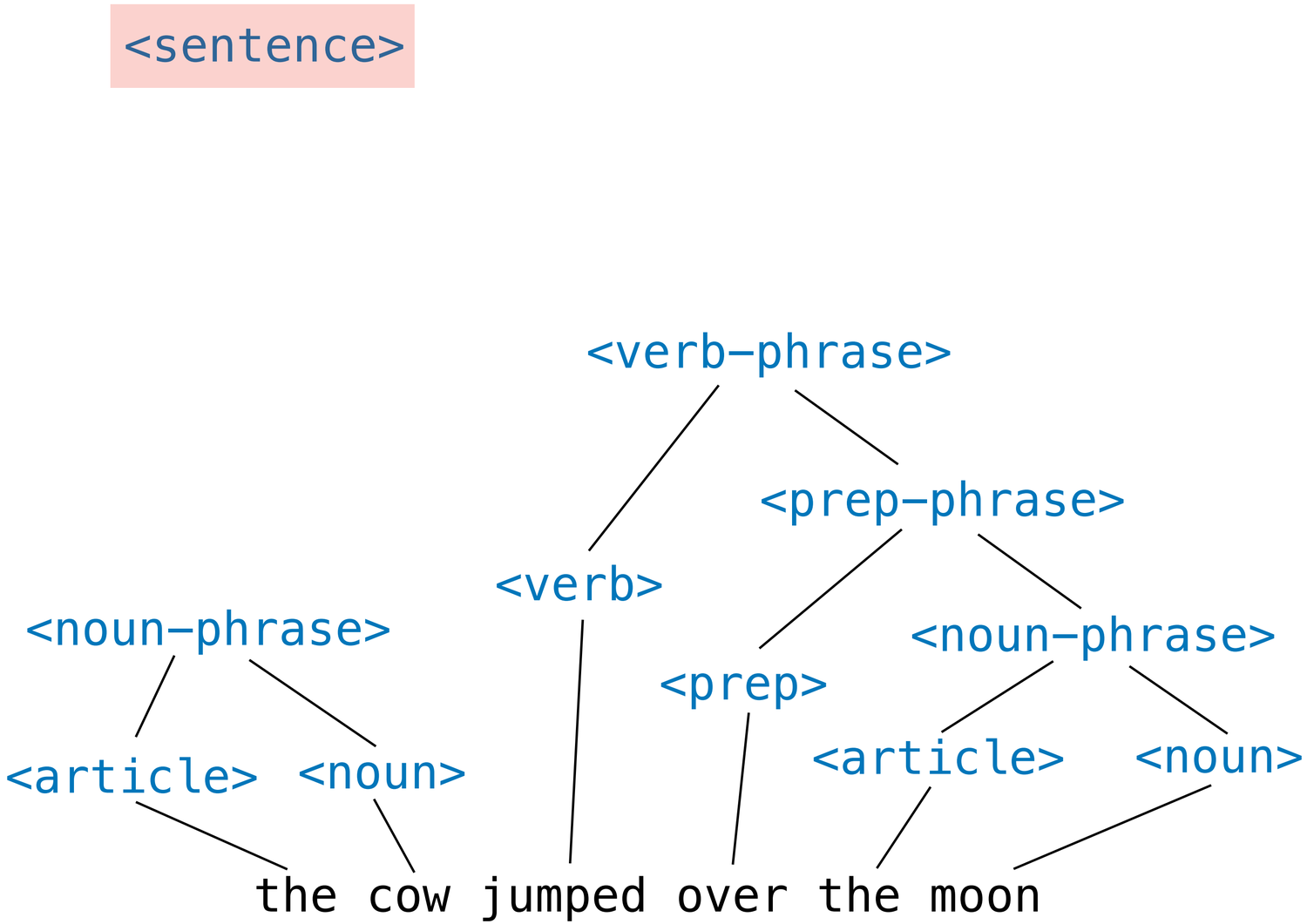

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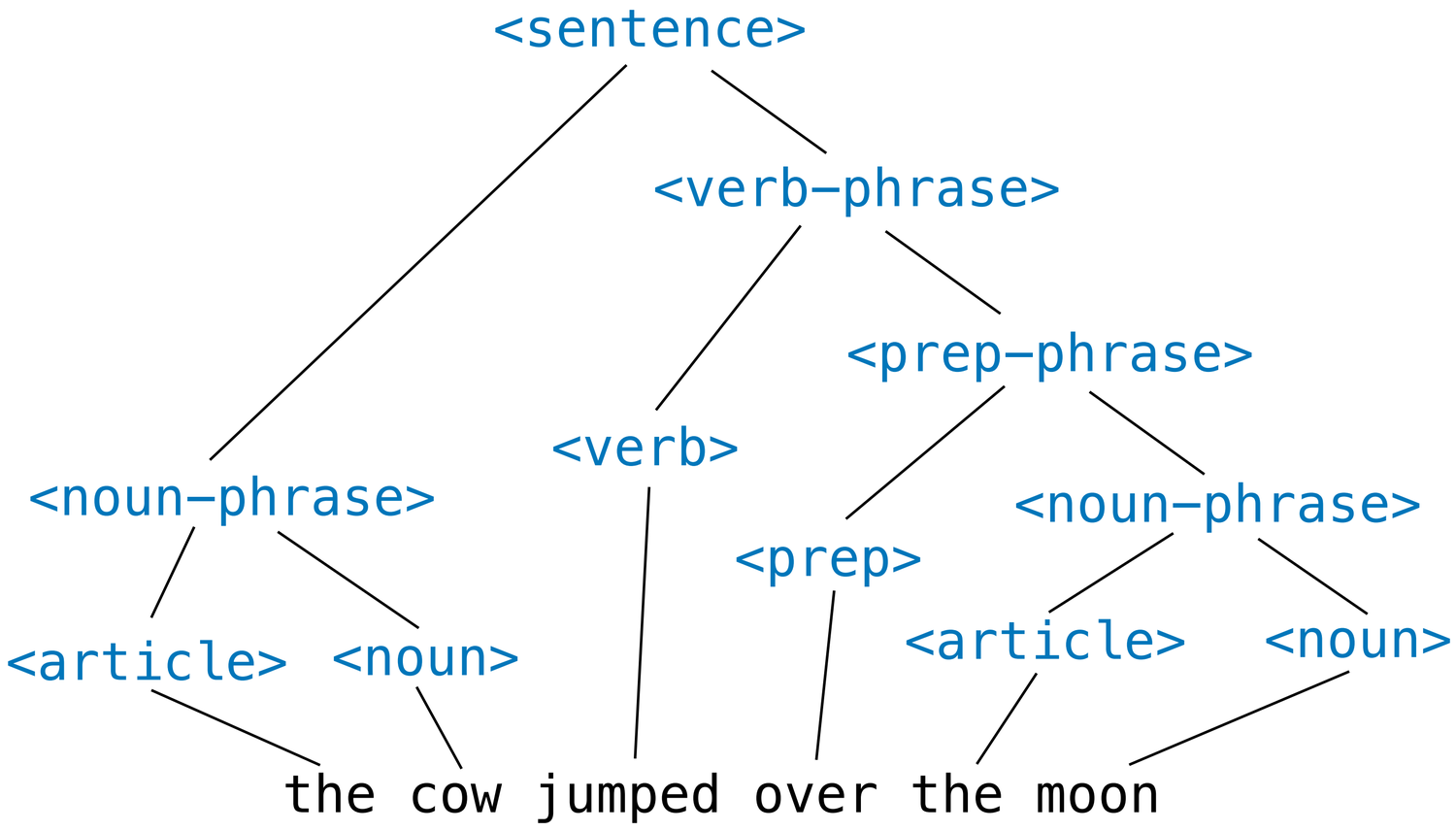
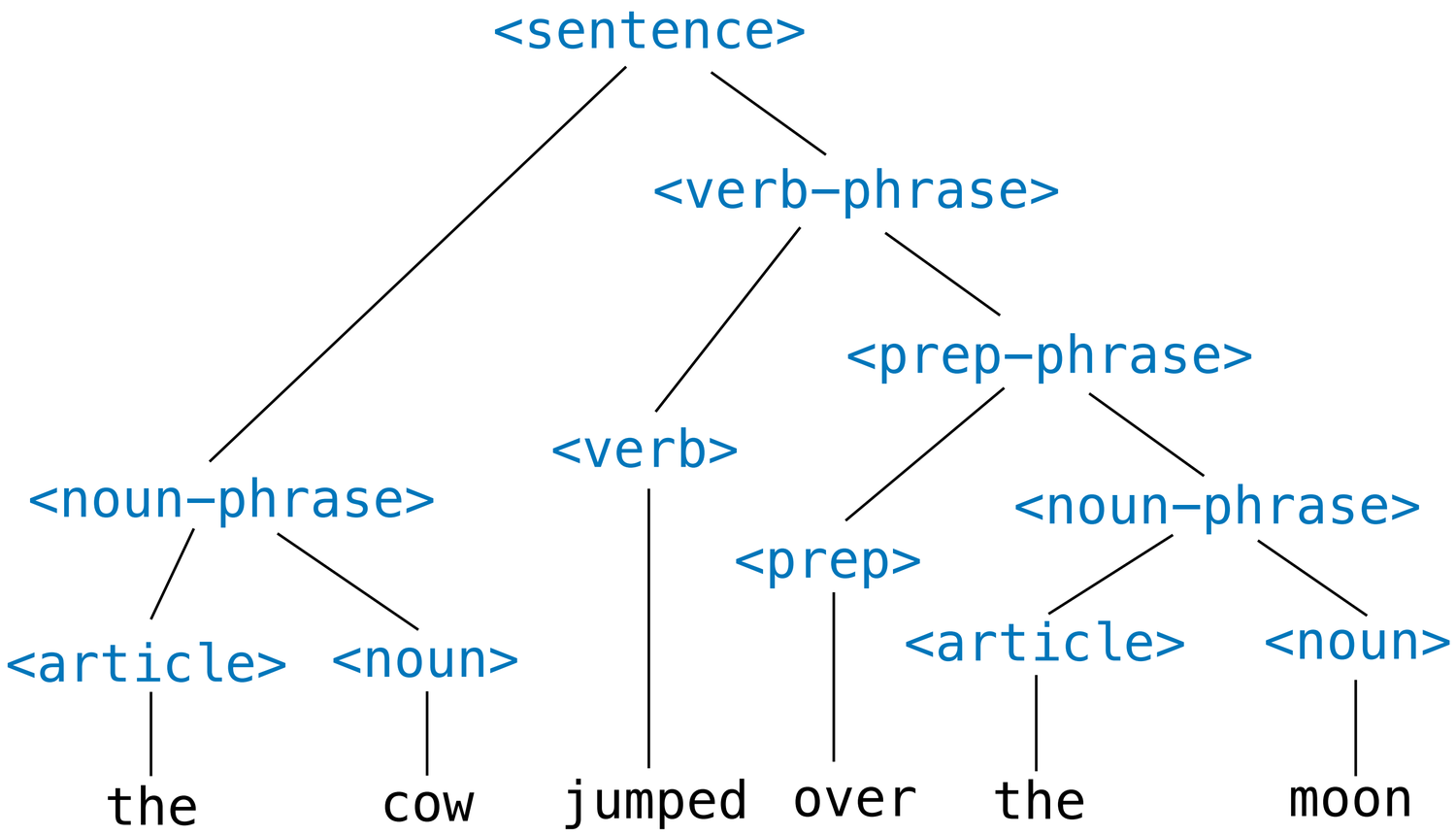


A Parse Tree



A derivation encodes hierarchical structure

Definitions (Symbols and Sentences)

<noun-phrase> jumped over <noun-phrase>

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Nonterminal symbols are symbols we will be allowed to "expand" (e.g., **<article>**)

Definitions (Symbols and Sentences)

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A **grammar** is define in terms of a collection of symbols

Nonterminal symbols are symbols we will be allowed to "expand" (e.g., **<article>**)

Terminal symbols are symbols cannot be further expanded (e.g. *moon*)

Definitions (Symbols and Sentences)

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A **grammar** is define in terms of a collection of symbols

Nonterminal symbols are symbols we will be allowed to "expand" (e.g., **<article>**)

Terminal symbols are symbols cannot be further expanded (e.g. *moon*)

A **sentential form** is a sequence of terminal or nonterminal symbols

Definitions (Symbols and Sentences)

<noun-phrase> jumped over <noun-phrase>

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A **sentence** is a sequence of *only terminal* symbols

Production Rules

```
<non-term> ::= sent-form1 | sent-form2 | ...
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A **(BNF) production rule** describes what we can replace a non-terminal symbol with in a derivation

The "|" means: we can replace it with one or the other sentential forms on either side of the "|"

Recall: Let-Expressions (Syntax Rule)

```
<expr> ::= let <var> = <expr> in <expr>
```

If x is a valid variable name, and e_1 is a well-formed expression and e_2 is a well-formed expression then

$$\texttt{let}\ x\ \texttt{=}\ e_1\ \texttt{in}\ e_2$$

is a well-formed expression

Recall: Let-Expressions (Syntax Rule)

```
<expr> ::= let <var> = <expr> in <expr>
```

Grammar for Thought Experiment

```
<sentence>     ::= <noun-phrase> <verb-phrase>

<verb-phrase> ::= <verb> <prep-phrase>

<noun>         ::= cow | moon
```

BNF Grammar

```
<sentence>    ::= <noun-phrase> <verb-phrase>
<verb-phrase> ::= <verb> <prep-phrase>
                | <verb>
<prep-phrase> ::= <prep> <noun-phrase>
<noun-phrase> ::= <article> <noun>
<article>     ::= the
<noun>        ::= cow
                | moon
<verb>        ::= jumped
<prep>        ::= over
```

BNF Grammar

A **BNF grammar** is defined by a collection of production rules and a **starting (nonterminal) symbol**

```
<sentence>    ::= <noun-phrase> <verb-phrase>
<verb-phrase> ::= <verb> <prep-phrase>
                | <verb>
<prep-phrase> ::= <prep> <noun-phrase>
<noun-phrase> ::= <article> <noun>
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Note. We don't specify the symbols of a grammar, they are implicit in the rules

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BNF Grammar

A **BNF grammar** is defined by a collection of production rules and a **starting (nonterminal) symbol**

Note. We don't specify the symbols of a grammar, they are implicit in the rules

Note. We don't specify the start symbol, it's the left nonterminal symbol in the **first rule**

```
<sentence>    ::= <noun-phrase> <verb-phrase>
<verb-phrase> ::= <verb> <prep-phrase>
                | <verb>
<prep-phrase> ::= <prep> <noun-phrase>
<noun-phrase> ::= <article> <noun>
<article>     ::= the
<noun>        ::= cow
                | moon
<verb>        ::= jumped
<prep>        ::= over
```

Another Example

```
<expr>   ::= <op1> <expr>
           | <op2> <expr> <expr>
           | <var>
<op1>    ::= not
<op2>    ::= and | or
<var>    ::= x | y | z
```

and x y

Another Example

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<expr>   ::= <op1> <expr>
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tokens (terminal symbols)

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abstractions (non-terminal symbols)

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production rules

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abstractions (non-terminal symbols)

tokens (terminal symbols)

Derivations and Parse Trees

Derivations and Parse Trees

Definition. A **derivation** is a sequence of sentential forms (beginning at the start symbol) in which each form is the result of **replacing <u>one</u> non-terminal symbol in the previous form** according to a production rule

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```
<expr>
<op2> <expr> <expr>
and <expr> <expr>
and <op1> <expr> <expr>
and not <expr> <expr>
and not <var> <expr>
and not x <expr>
and not x <var>
and not x y
```

Derivations and Parse Trees

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<expr>  ::= <op1> <expr>
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<expr>

<expr>

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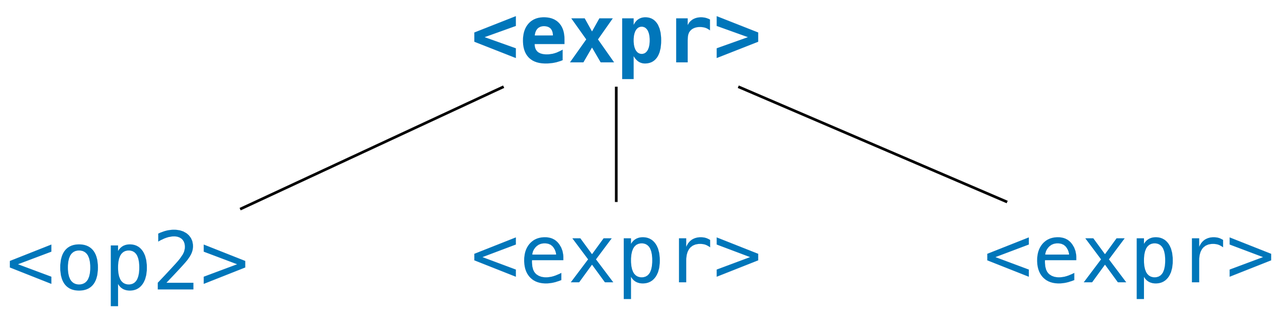
```

```
<expr>
<op2> <expr> <expr>
```



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<expr>  ::= <op1> <expr>
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```

```
<expr>
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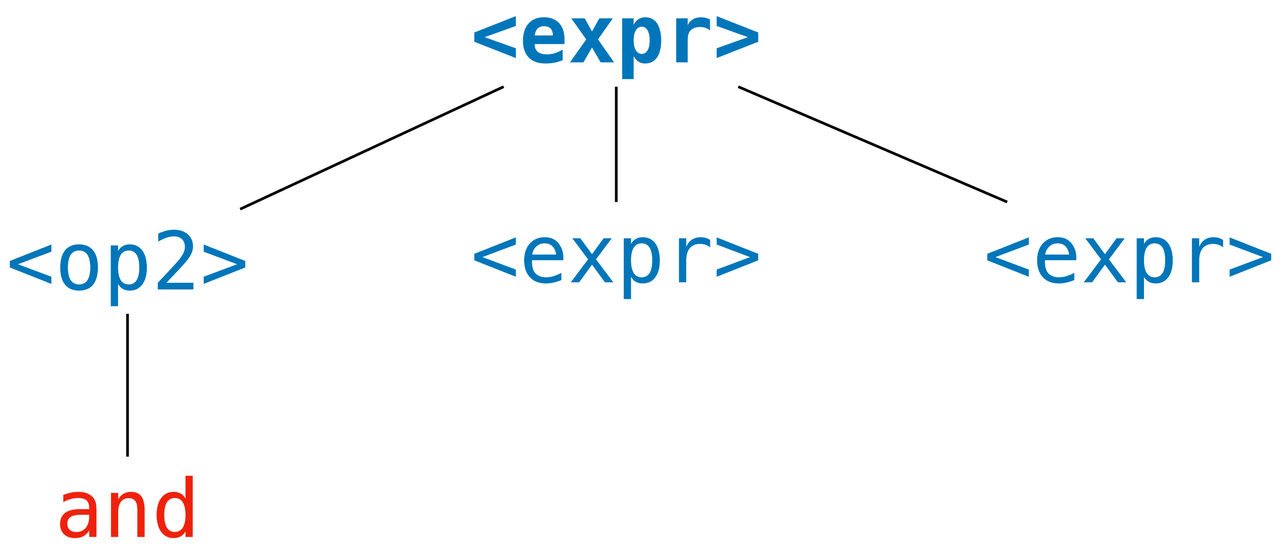

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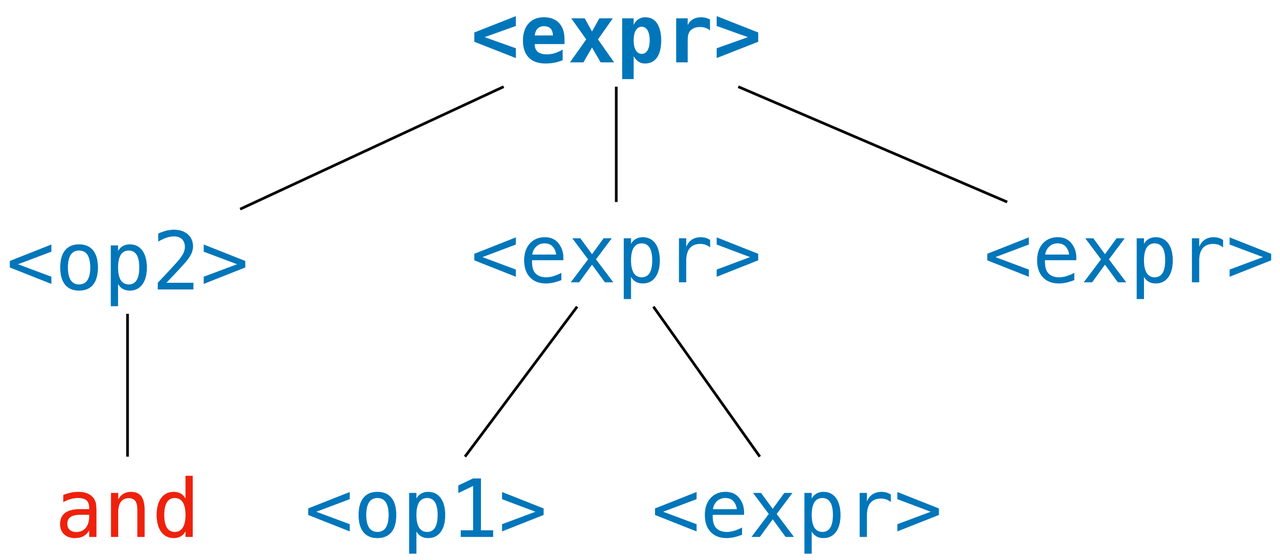

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```
<expr>
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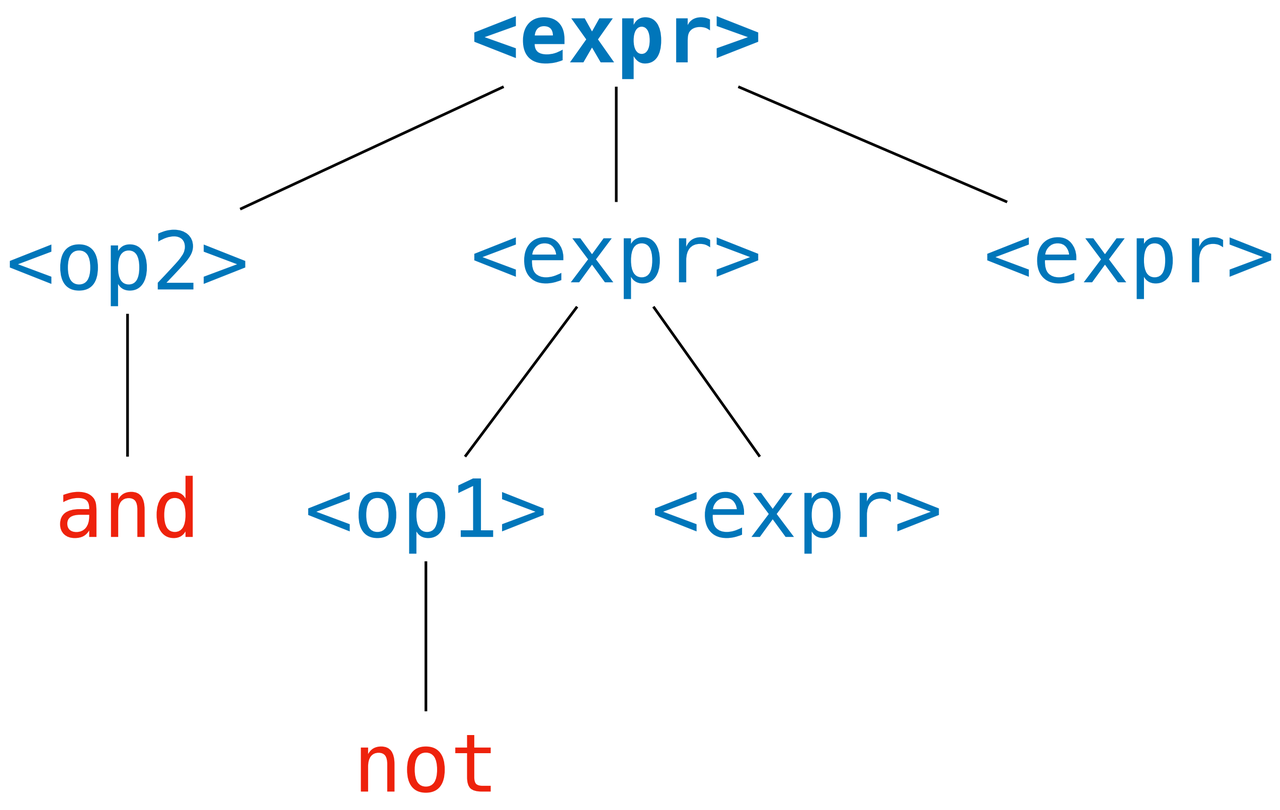

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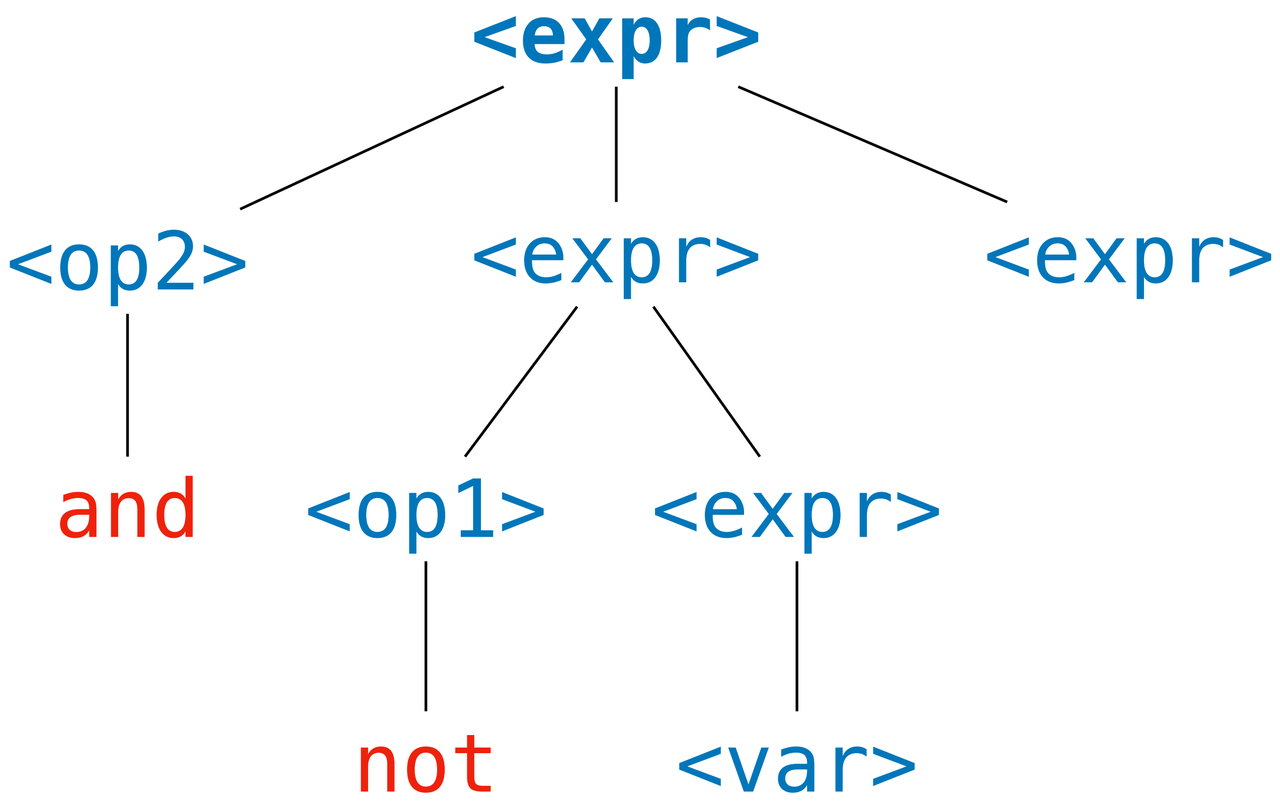

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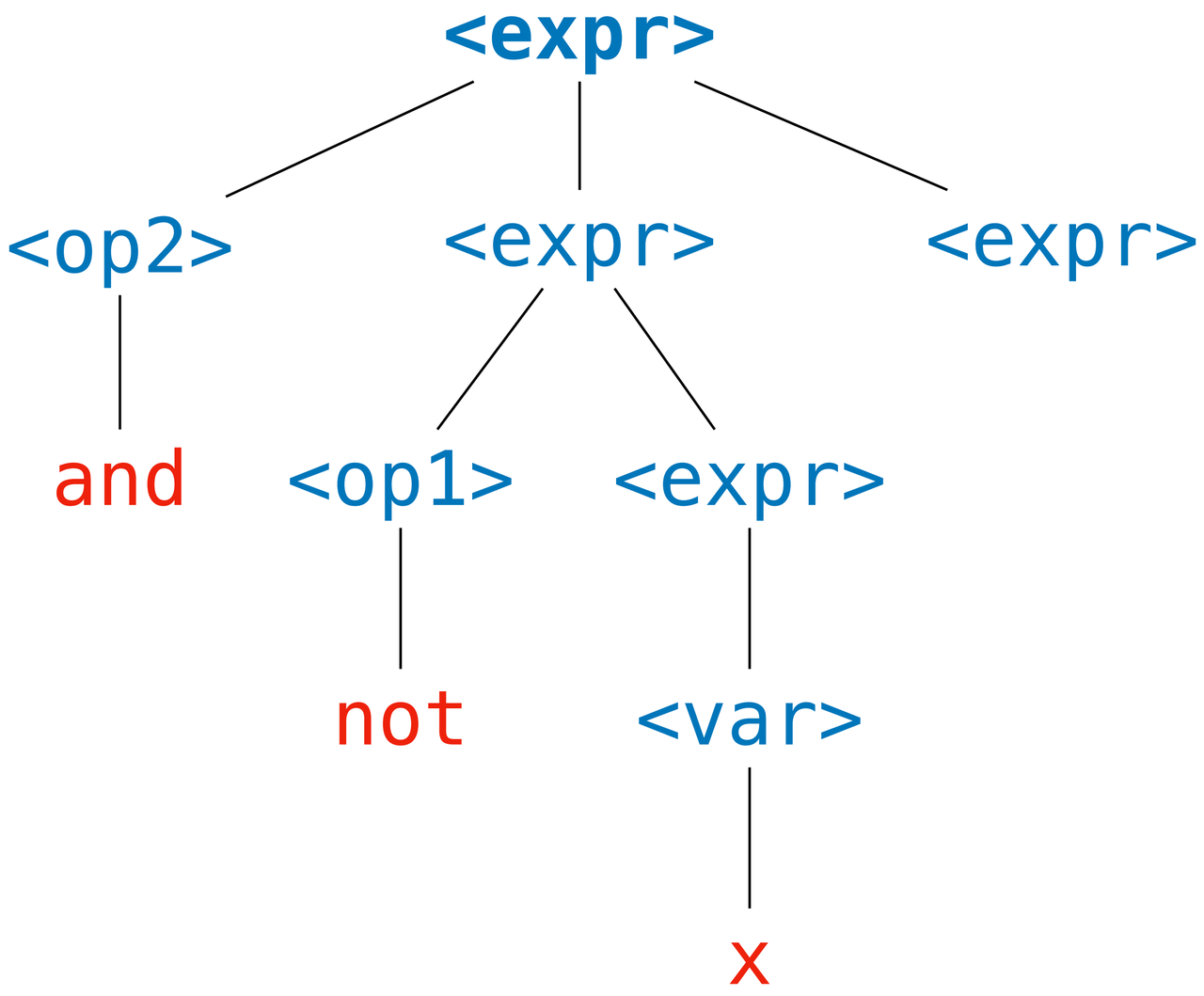
```



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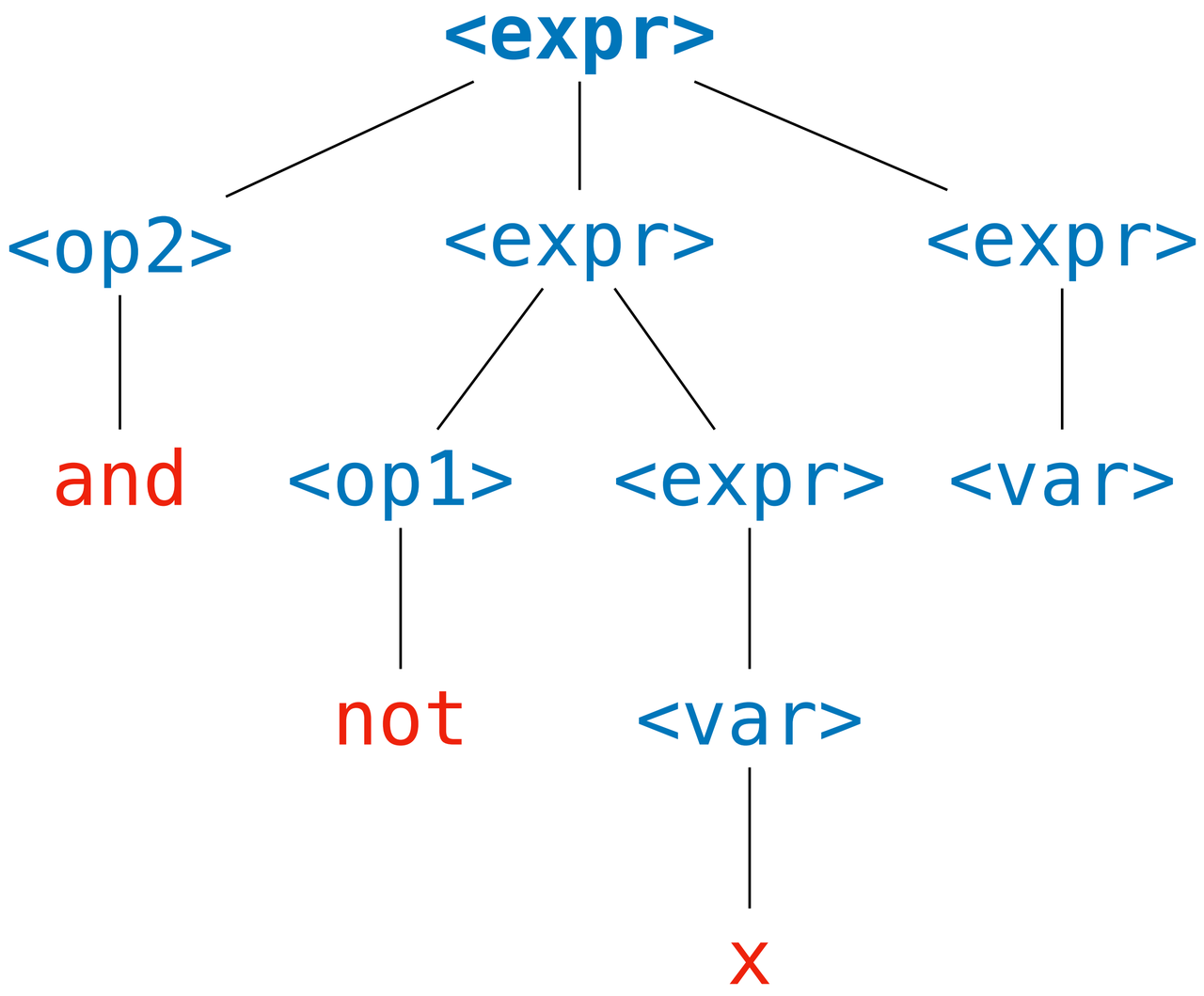

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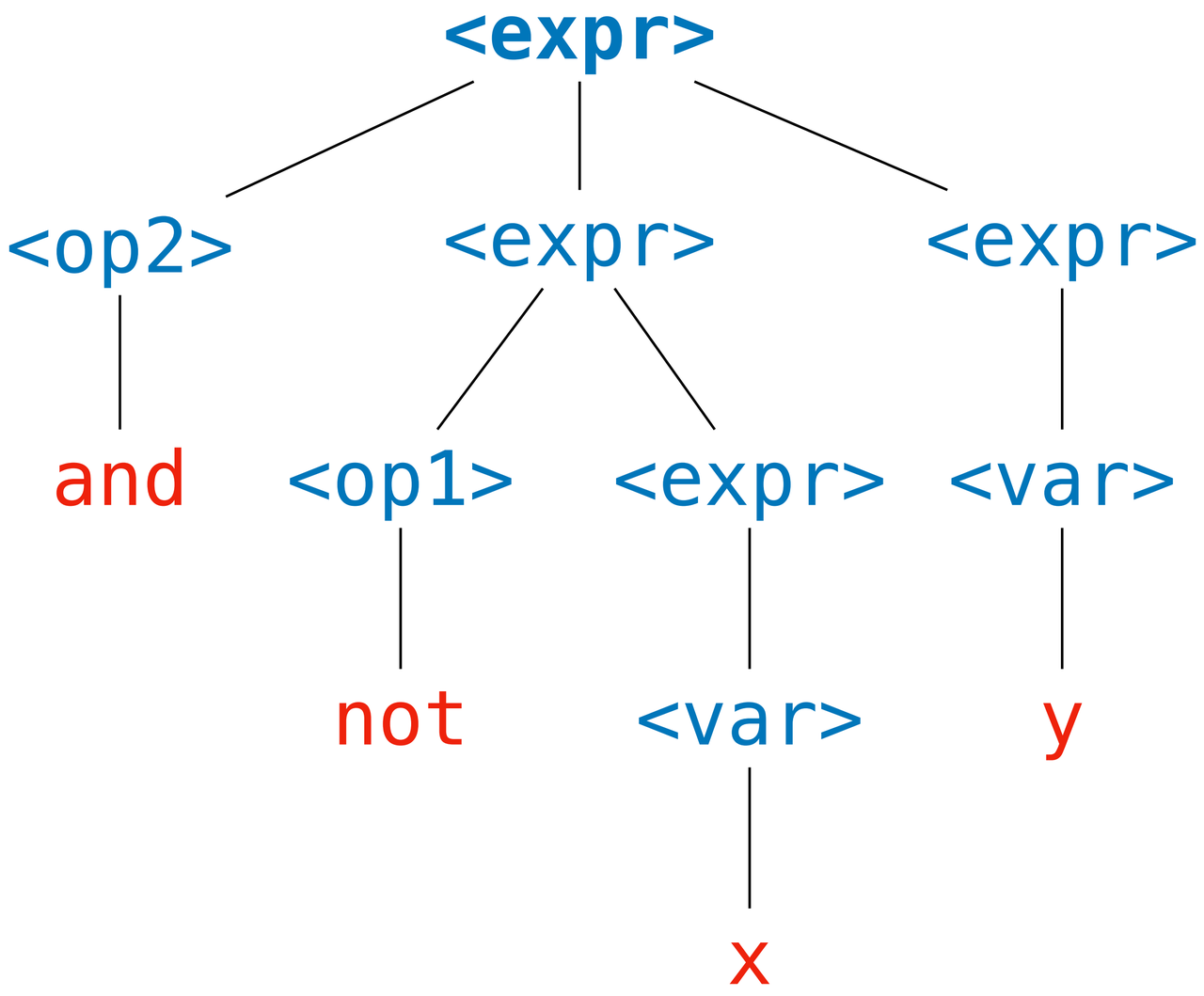

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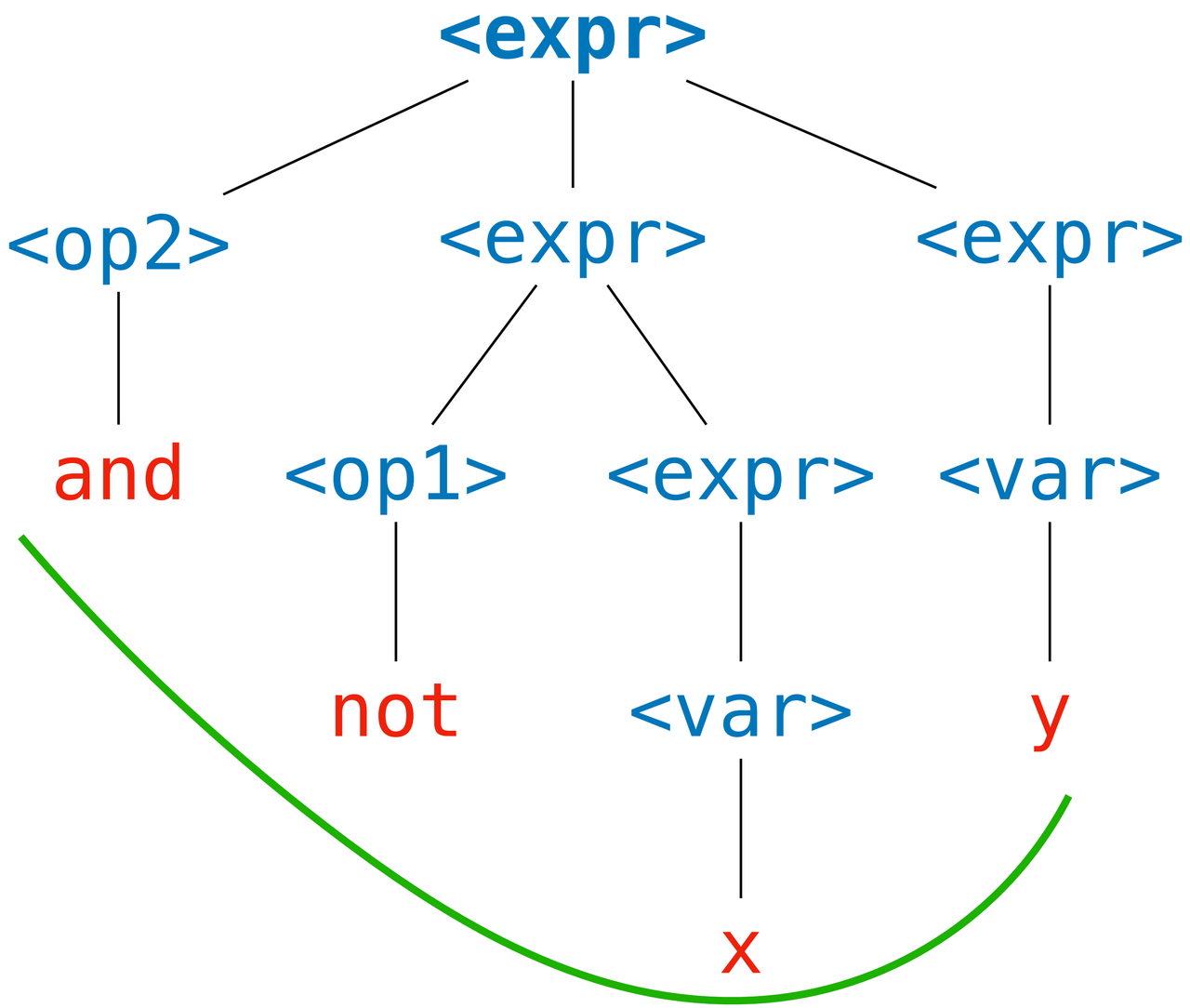

The point: parse trees and derivations represent the same hierarchical structure

Why do we care?

Why do we care?

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Why do we care?

We will parse **token streams** into **parse trees**

Why do we care?

We will parse **token streams** into **parse trees**

It is much easier to evaluate something hierarchical than something which is linear

Practice Problem

```
<expr>  ::= <op1> <expr>
          | <expr> <op2> <expr>
          | <var>
<op1>   ::= not
<op2>   ::= and | or
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```

*Give a derivation of **not x and y or z** in the above grammar, both as a sequence of sentential forms and as a parse tree*

*(In Python, if **x** and **y** and **z** are **True**, what does this expression evaluate to?)*

Answer

```
<expr>   ::= <op1> <expr>
           | <expr> <op2> <expr>
           | <var>
<op1>    ::= not
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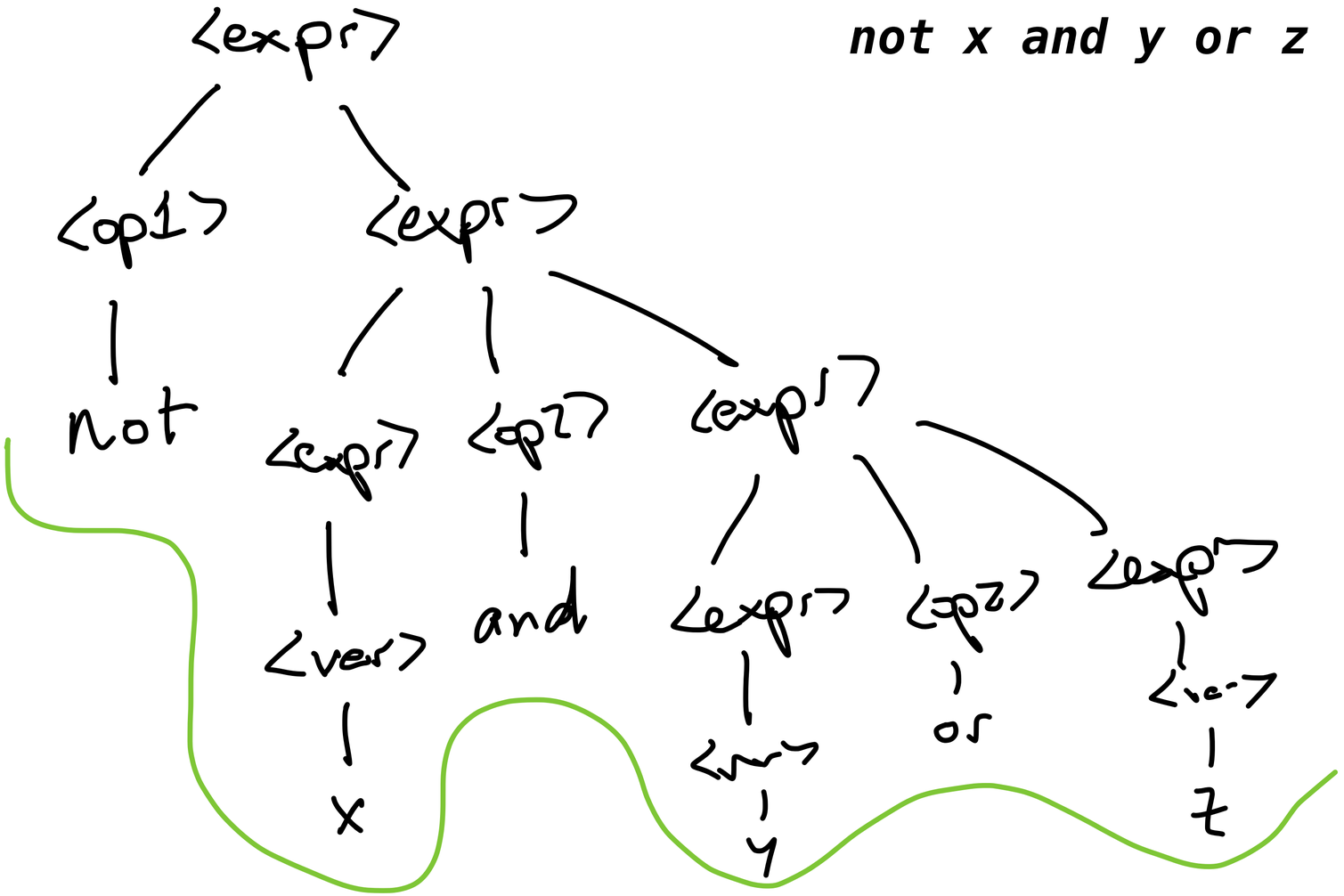
```

not x and y or z



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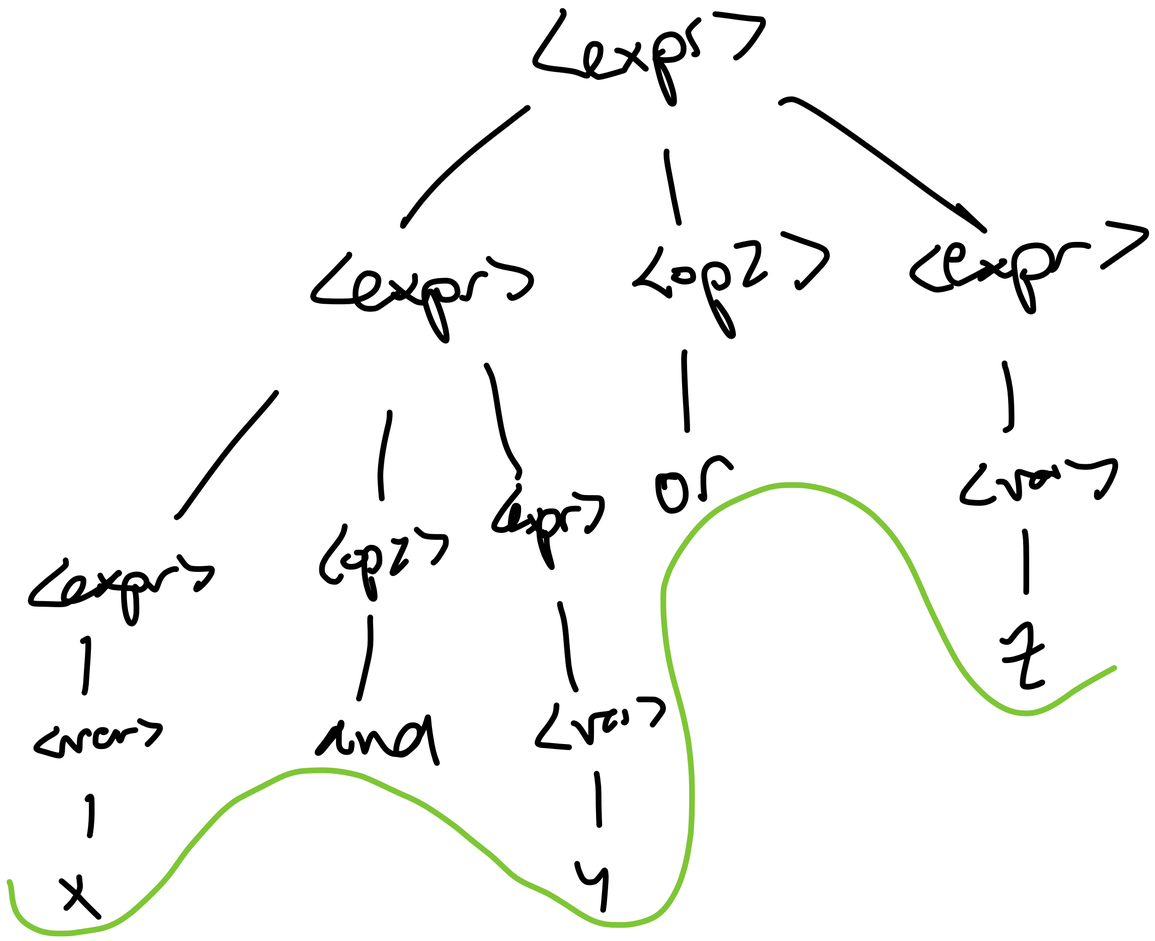

An Example from 320Caml

```
let x = 2 in if x = z then x else y
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How can we demonstrate that this is a well-formed expression?

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Answer: Well build a derivation/parse tree for it with the root **<expr>**!

Recall: Let-Expressions (Syntax Rule)

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```

If x is a valid variable name, and e_1 is a well-formed expression and e_2 is a well-formed expression then

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Recall: If-Expressions (Syntax Rule)

```
<expr> ::= if <expr> then <expr> else <expr>
```

If e_1 is a well-formed expression and e_2 is a well-formed expression and e_3 is a well-formed expression, then

$$\texttt{if}\ e_1\ \texttt{then}\ e_2\ \texttt{else}\ e_3$$

is a well-formed expression

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Ambiguity

Ambiguity in Natural Language

The duck is ready for dinner.

John saw the man on the mountain with a telescope.

He said on Tuesday there would be an exam.

(examples taken from around the internet)

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We have informal tactics for avoiding these pitfalls

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Ambiguity in Natural Language

*The duck is ready **to eat** dinner.*

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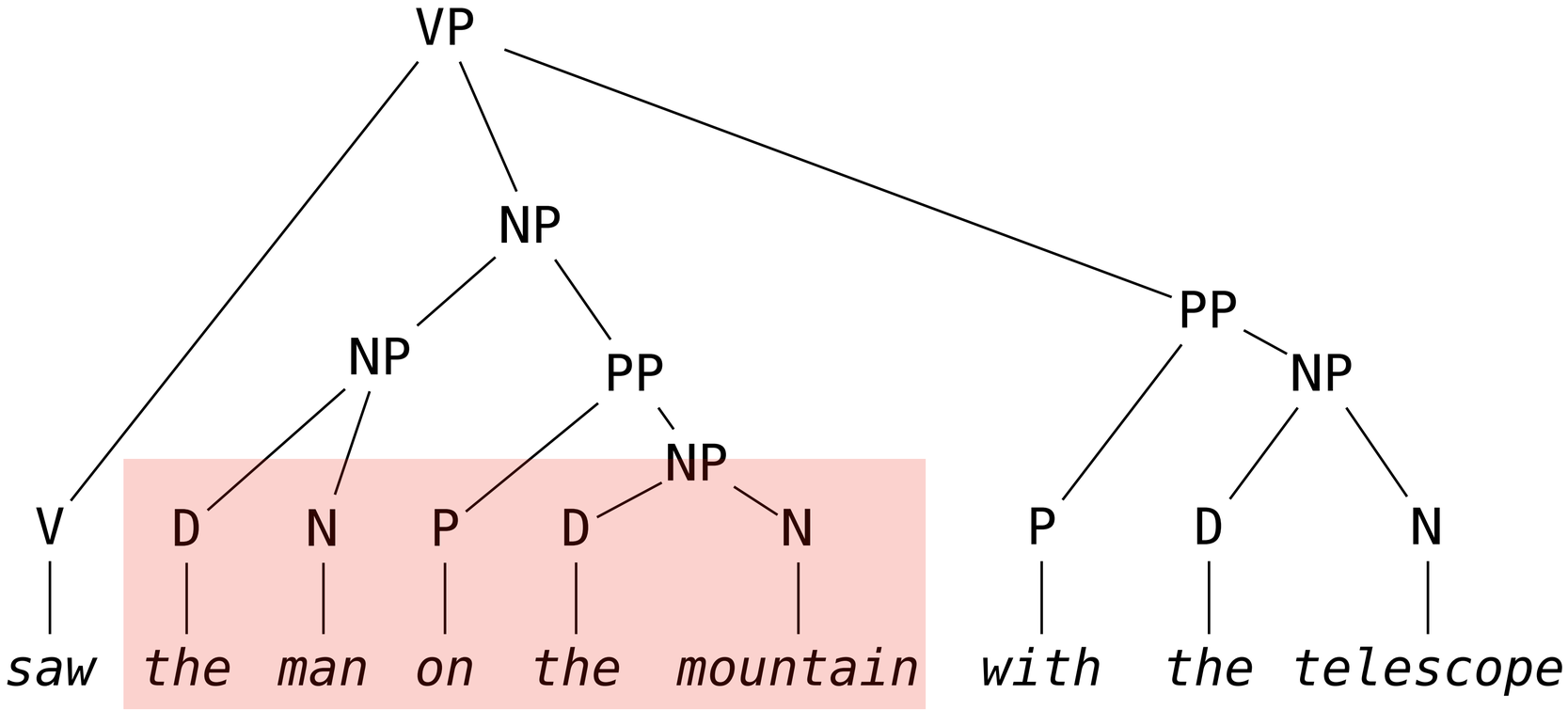
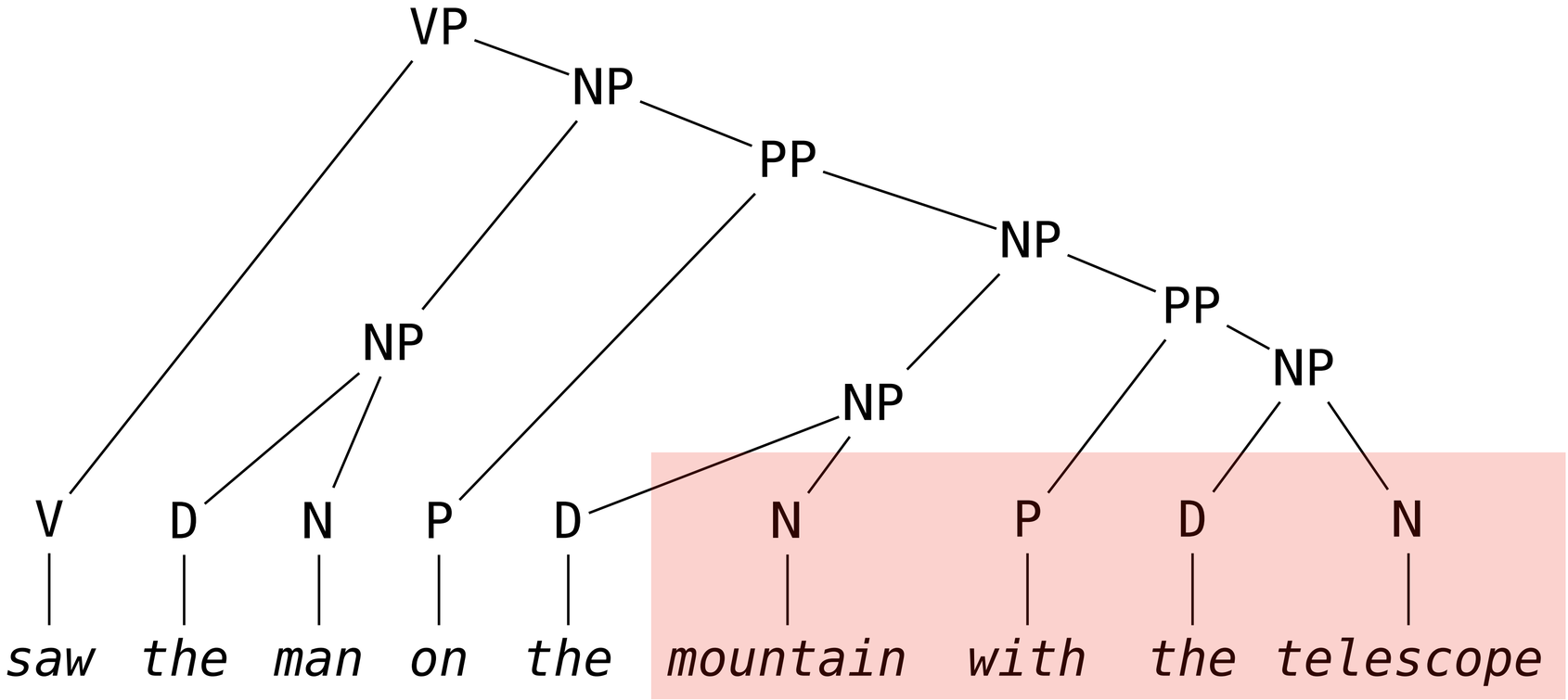
Aside: Ambiguity and Linearity



Ambiguity is caused by writing down hierarchical structures in a linear fashion

Aside: Ambiguity and Linearity



If we know the parse tree, then there is no ambiguity

The hierarchical structure determines the meaning of the sentence

Ambiguity in Formal Grammar

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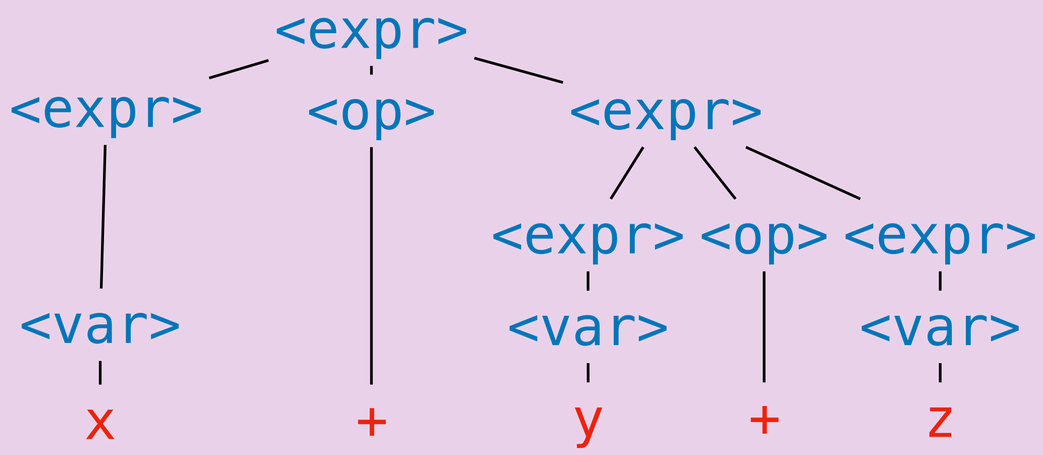

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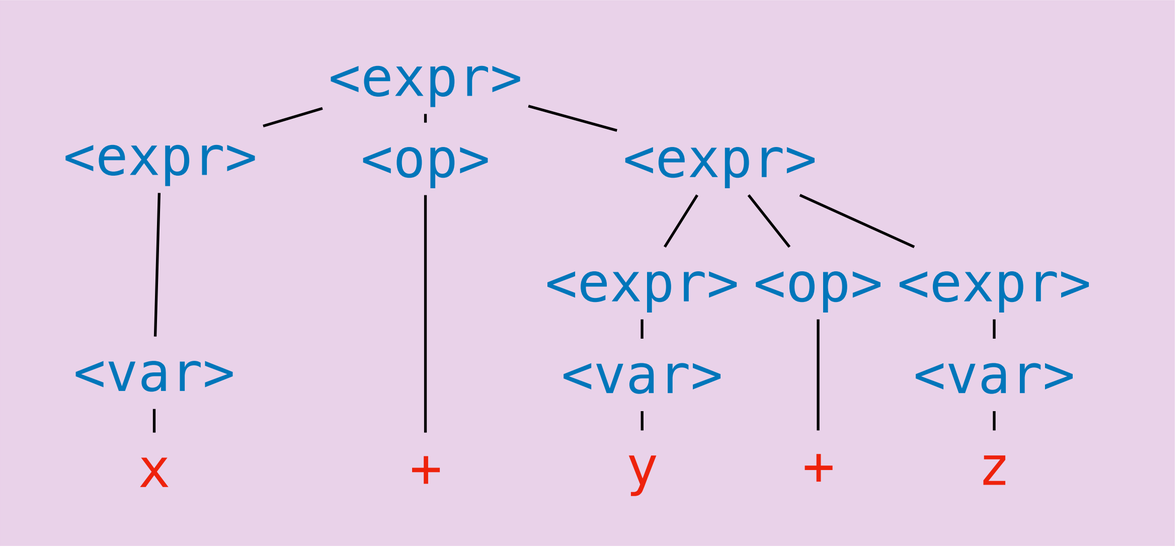
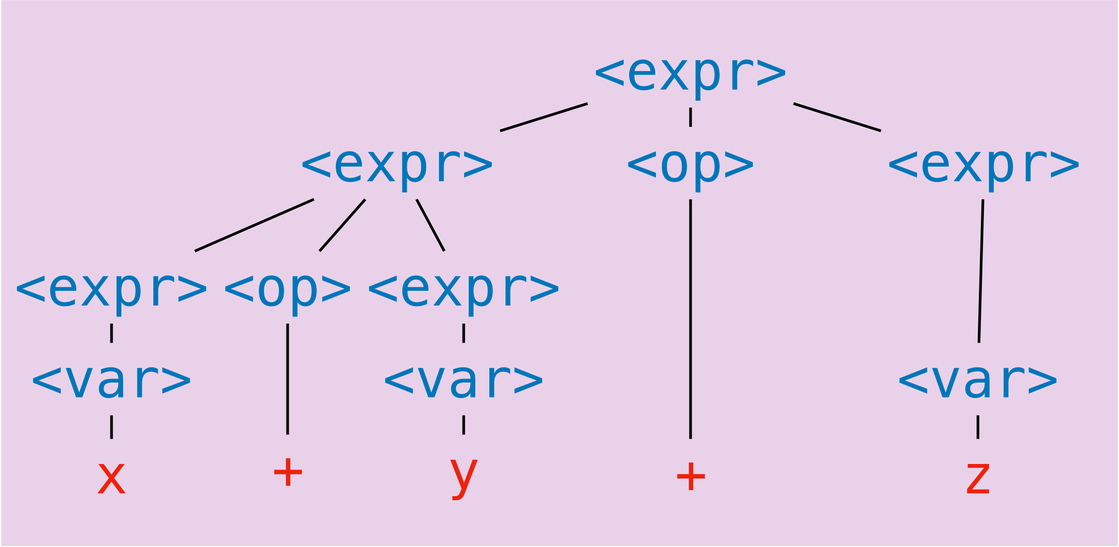
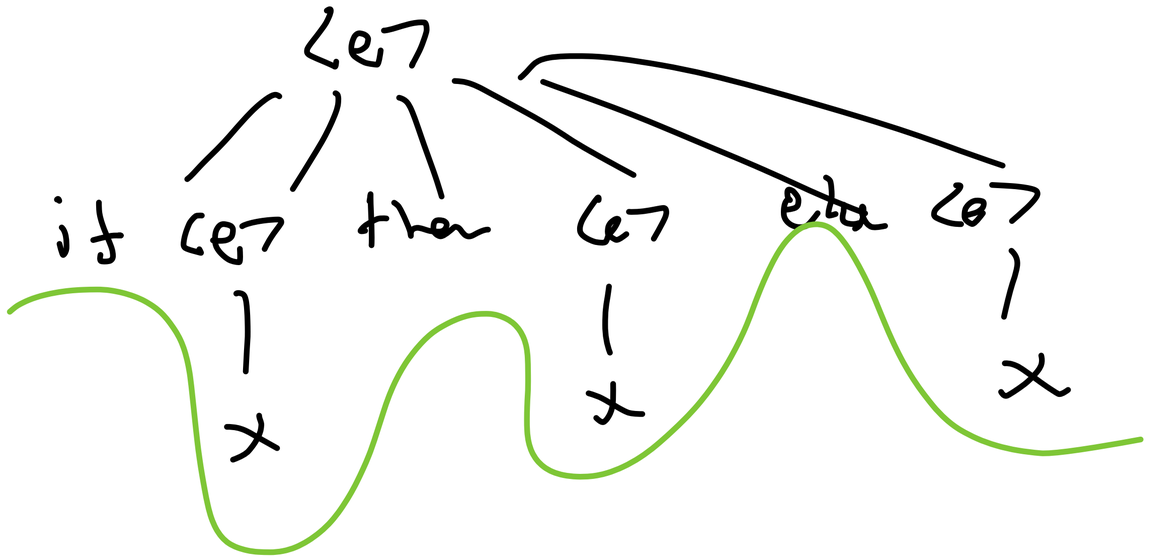




Example

```
<expr> ::= x
         | if <expr> then <expr>
         | if <expr> then <expr> else <expr>
```

if x then x else x



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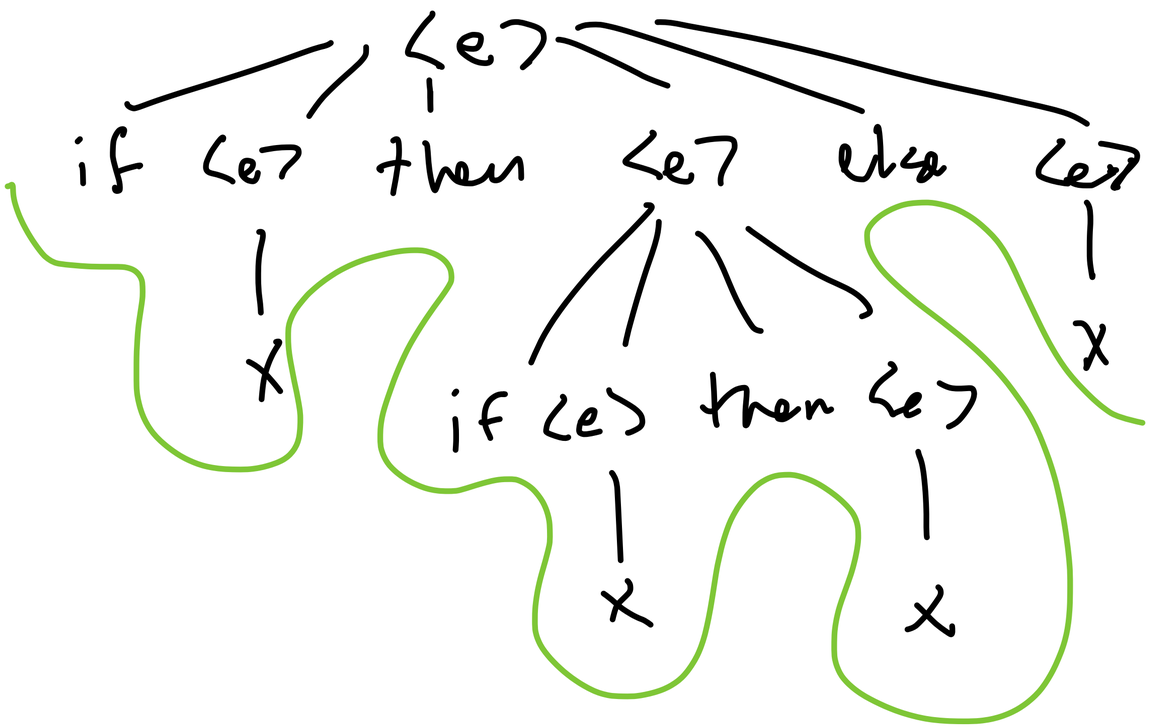

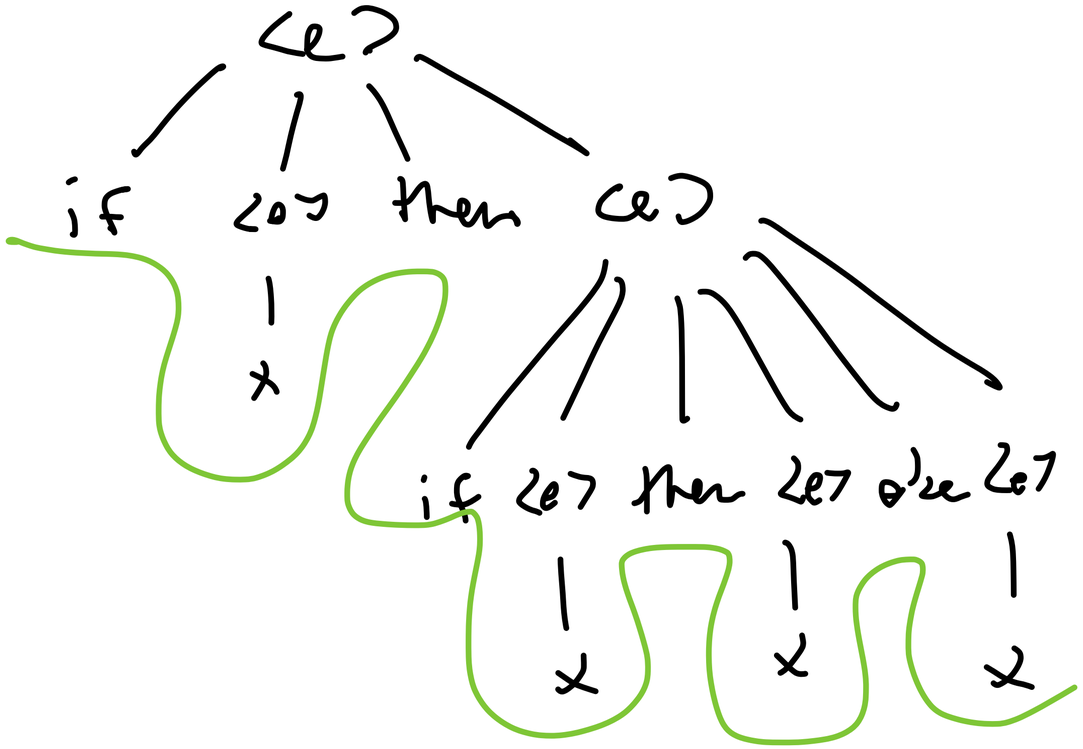

What can we do about ambiguity?

Fixity

Fixity

The <u>fixity</u> of an operator refers to where an operator is placed relative to its arguments:

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prefix `f x , (- x)`

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prefix	`f x , (- x)`
postfix	`a!`
infix	`a * b, a + b, a mod b`

Fixity

The fixity of an operator refers to where an operator is placed relative to its arguments:

prefix	`f x , (- x)`
postfix	`a!`
infix	`a * b, a + b, a mod b`
mixfix	`if b then x else y`

Polish Notation

```
- / + 2 * 1 - 2 3
```

is equivalent to

```
- (2 + (1 * (- 2) / 3)
```

To avoid ambiguity, we can make all operators prefix (or postfix) operators. *We don't even need parentheses*

(This how early calculators worked)

Example

```
<expr> ::= <bool>
         | <var>
         | ifthen <expr> <expr>
         | ifthenelse <expr> <expr> <expr>
<bool>::= tru | fls
<var> ::= x | y | z
```

No more ambiguity. But programs written like this are notoriously difficult to read...

Lots of Parentheses

```
<expr>  ::= (<op1> <expr>)
          | (<expr> <op2> <expr>)
          | <var>
<op1>   ::= not
<op2>   ::= and | or
<var>   ::= x | y | z
```

Lots of Parentheses

```
<expr>  ::= (<op1> <expr>)
          | (<expr> <op2> <expr>)
          | <var>
<op1>   ::= not
<op2>   ::= and | or
<var>   ::= x | y | z
```

If we want infix operators, we *could* add parentheses around all operators

Lots of Parentheses

```
<expr>  ::= (<op1> <expr>)
          | (<expr> <op2> <expr>)
          | <var>
<op1>   ::= not
<op2>   ::= and | or
<var>   ::= x | y | z
```

```
(((not x) and (not (not y)) or z)

((((x or y) or z) or x) or y)

(not ((not x) and (not y)) or
      (x and z))

(x and y)
```

If we want infix operators, we *could* add parentheses around all operators

But we run into a similar issue: *Too many parentheses are difficult to read*

Can we get away without (or with fewer) parentheses?

Two Properties of Infix operators

Two Properties of Infix operators

Associativity: *How should arguments be grouped in an expression like* `1 + 2 + 3 + 4`*?*

$$(((1 - 2) - 3) - 4) - 5$$

Precedence: *How should arguments be grouped in an expression like* `1 + 2 * 3 + 4`*?*

Associativity

The **associativity** of an infix operator refers to how its arguments are grouped in the absence of parentheses:

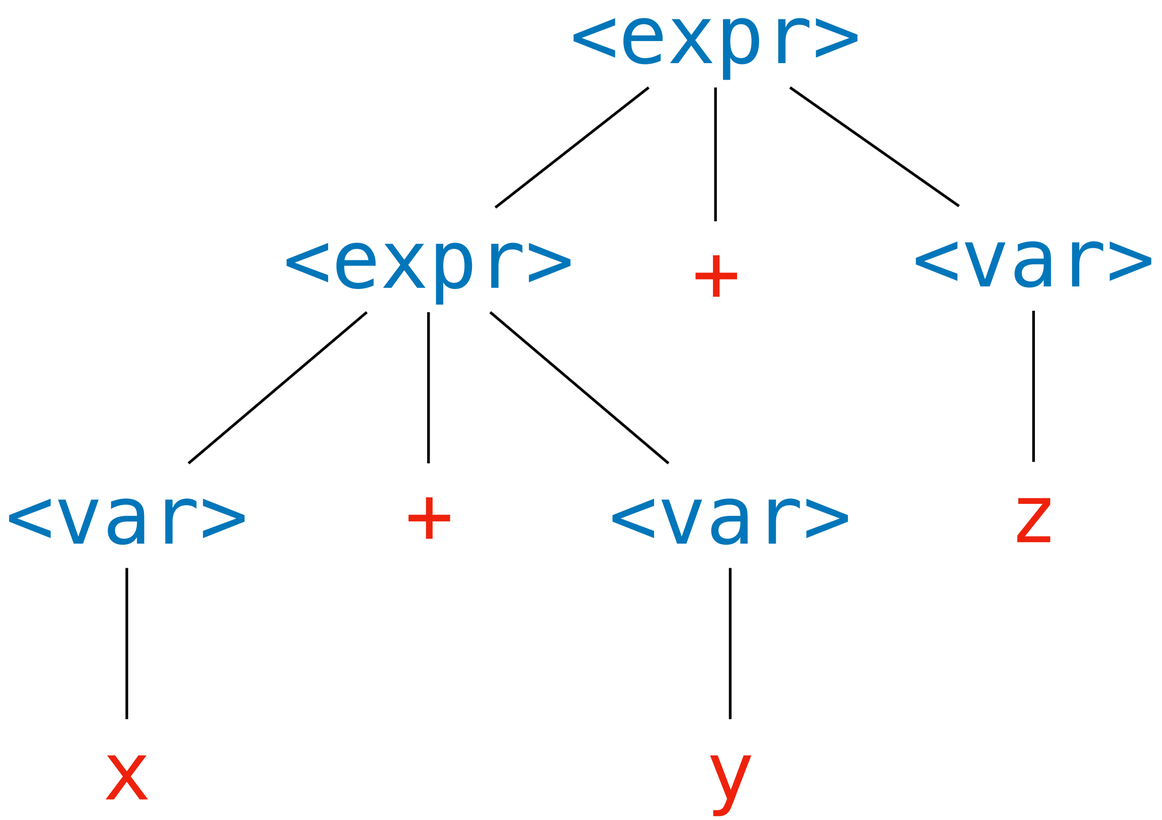
left associative `1 + 2 + 3` ⇒ `(1 + 2) + 3`

right associative `a -> b -> c` ⇒ `a -> (b -> c)`

Associativity

```
<expr> ::= <expr> + <expr>
         | <var>
<var>  ::= x | y | z
```

x + y + z ⇒



"*add the sum of x and y to z*"

How do we enforce that we get a tree of this shape?

The Culprit

```
<expr> ::= <expr> + <expr>
         | <var>
<var>  ::= x | y | z
```

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```
<expr> ::= <expr> + <expr>
         | <var>
<var>  ::= x | y | z
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Any time we have a rule like this, we should be suspicious...

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<expr> ::= <expr> + <expr>
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<expr> + <expr> ⇒ <expr> + <expr> + <expr>

The Culprit

```
<expr> ::= <expr> + <expr>
         | <var>
<var>  ::= x | y | z
```

Any time we have a rule like this, we should be suspicious...

<expr> + <expr> ⇒ <expr> + <expr> + <expr>

Which <expr> did we replace?

(((1 + 2) + 3) + 4) + 5

The Solution: Breaking Symmetry

```
<expr> ::= <expr> + <var>
         | <var>
<var>  ::= x | y | z
```

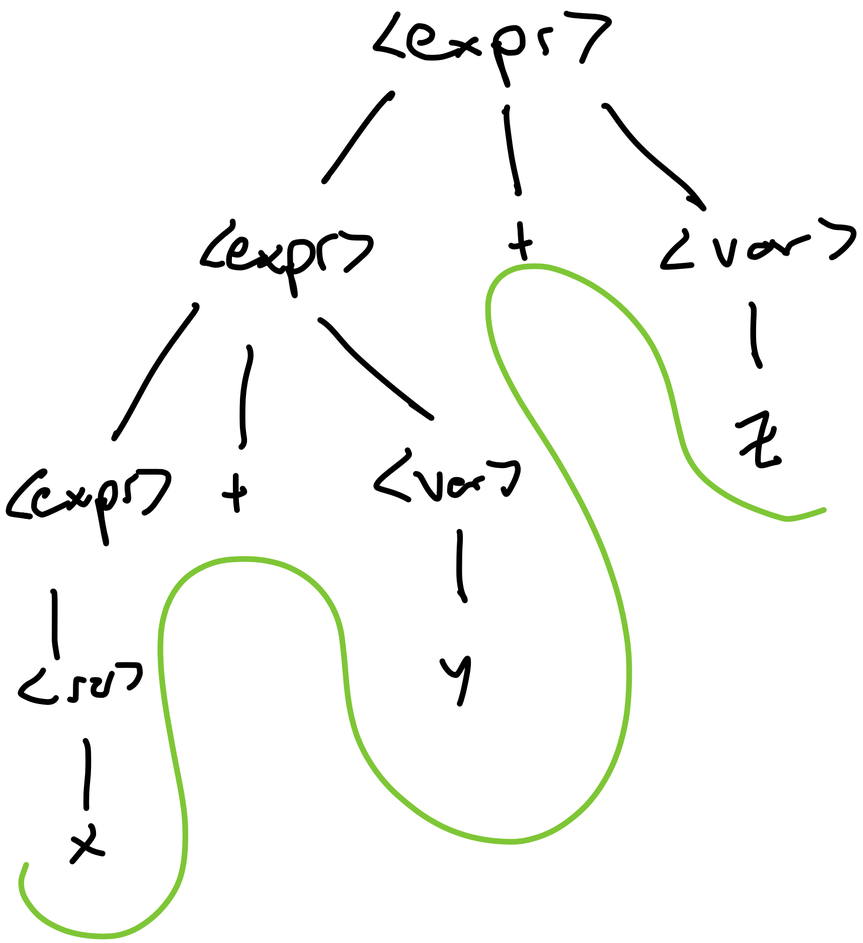
```
<expr>
<expr> + <var>
<expr> + z
<expr> + <var> + z
<expr> + y + z
<var> + y + z
x + y + z
```

We make sure that one of the arguments must be "simpler"

By enforcing that the second argument is a <var>, we'll get the left-associative parse tree

Example Parse Tree

```
<expr> ::= <expr> + <var>
         | <var>
<var>  ::= x | y | z
```




```
<expr>
<expr> + <var>
<expr> + z
<expr> + <var> + z
<expr> + y + z
<var> + y + z
x + y + z
```

And Right Associativity

```
<type> ::= <base> -> <type>
         | <base>
<base> ::= ()
```

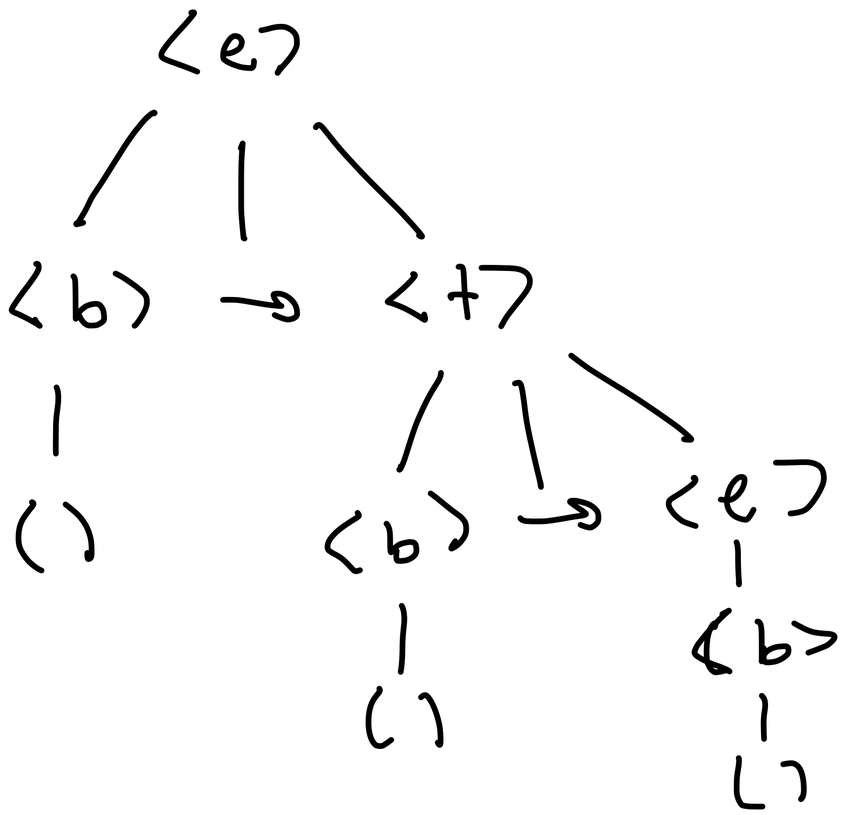
```
<type>
<base> -> <type>
() -> <type>
() -> <base> -> <type>
() -> () -> <type>
() -> () -> <base>
() -> () -> ()
```

For right associativity, we break symmetry by "factoring out" the *left* argument

Practice Problem

```
<type> ::= <base> -> <type>
         | <base>
<base> ::= ()
```

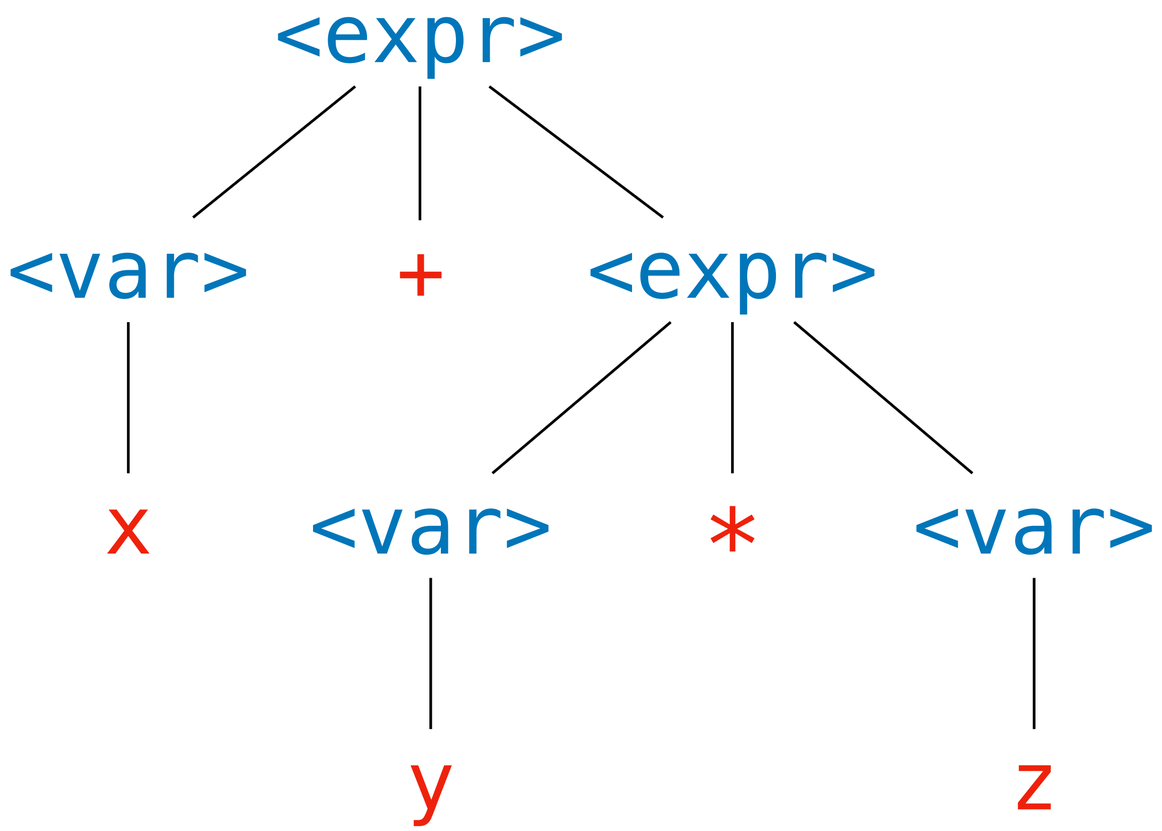
Draw the parse tree for `() -> () -> ()`



Multiple Operators

```
<expr> ::= <expr> <op> <var>
         | <var>
<op>   ::= + | *
<var>  ::= x | y | z
```

x + y * z ⇒



"add x to the product of y and z"

Question. *What if we have multiple operators? Which one should "bind tighter"?*

Precedence

$$2 + 3 \times 6 = 2 + (3 \times 6) = 20$$

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Example. **PEMDAS (paren, exp, mul, div, add, sub)**

Precedence

$$2 + 3 \times 6 = 2 + (3 \times 6) = 20$$

The <u>precedence</u> of an operator refers to order in which an operator should be considered, relative to other operators

Example. **PEMDAS (paren, exp, mul, div, add, sub)**

Higher precedence means it "binds tighter"

$\boxed{1} + \boxed{2 * 3} + \boxed{4 * 5 * 6} + \boxed{7} + \boxed{8}$

Dealing with Precedence within the Grammar

```
<expr> ::= <expr> + <term>
         | <term>
<term> ::= <term> * <var>
         | <var>
<var>  ::= x | y | z
```

We factor out the * part of the <expr> rule

Note that we handle *lower* precedence terms first, since terms *deeper* in the parse tree are evaluated first

A Note on Associativity and Precedence

```
%token PLUS
%token MINUS
%token TIMES
%token DIVIDE

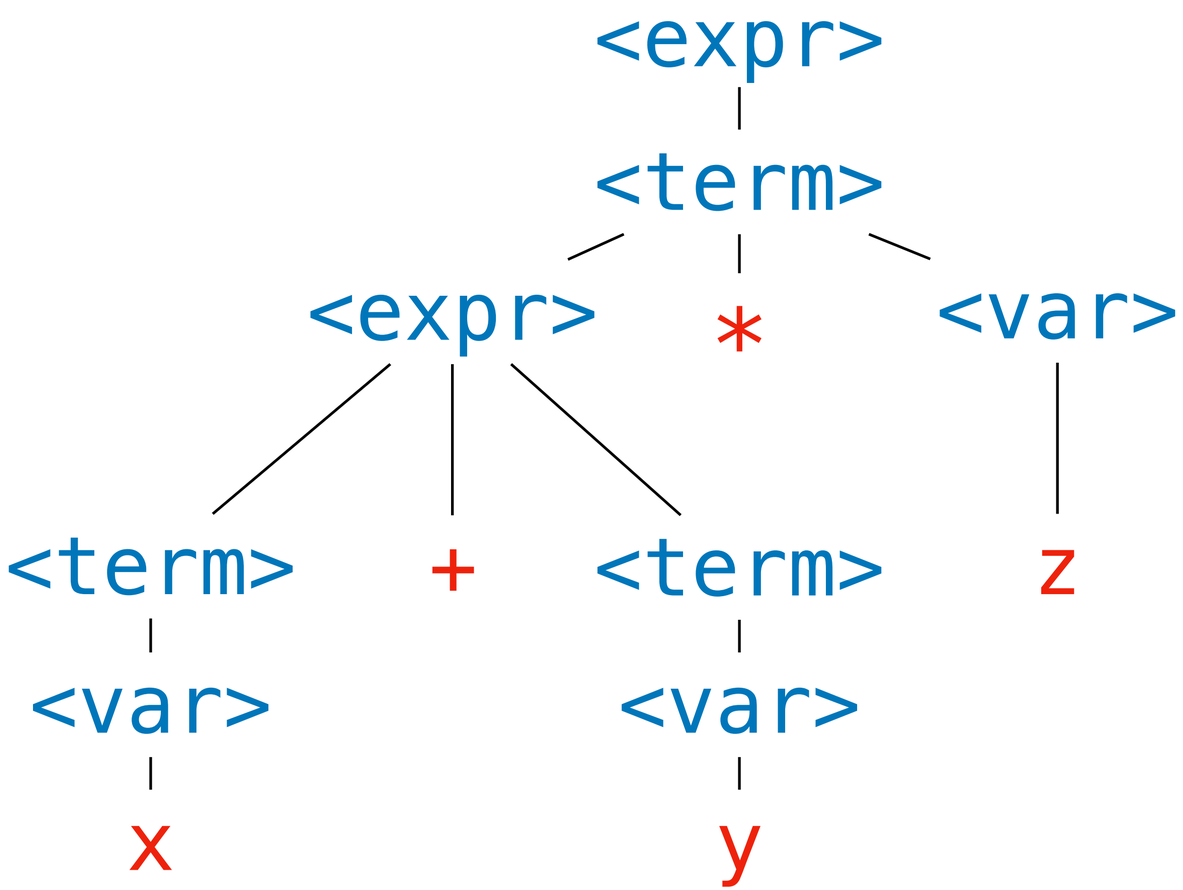
%left PLUS, MINUS
%left TIMES, DIVIDE
```

In most situations, we actually won't deal with associativity and precedence in this way

Using a parser generator we'll often be able to *specify* the precedence of an operator

The Issue of Parentheses Returns

```
<expr> ::= <expr> + <term>
         | <term>
<term> ::= <term> * <var>
         | <var>
<var>  ::= x | y | z
```



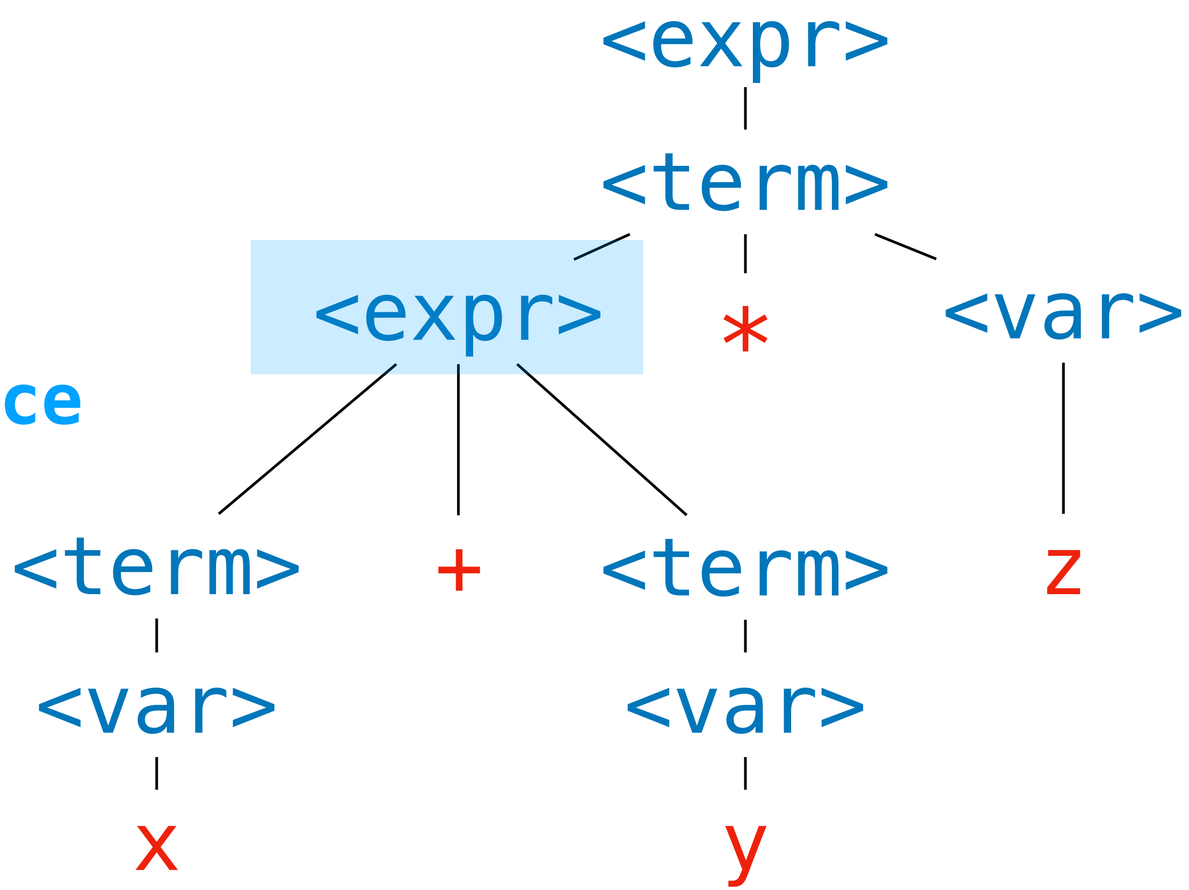
"*multiply the sum of x and y with z*"

Question. *Can we derive this parse tree?*

The Issue of Parentheses Returns

```
<expr> ::= <expr> + <term>
         | <term>
<term> ::= <term> * <var>
         | <var>
<var>  ::= x | y | z
```

No, we need to introduce parentheses again



"*multiply the sum of x and y with z*"

Question. *Can we derive this parse tree?*

Dealing with Parentheses

```
<expr> ::= <expr> + <term>
         | <term>
<term> ::= <term> * <pars>
         | <pars>
<pars> ::= <var> | ( <expr> )
<var>  ::= x | y | z
```

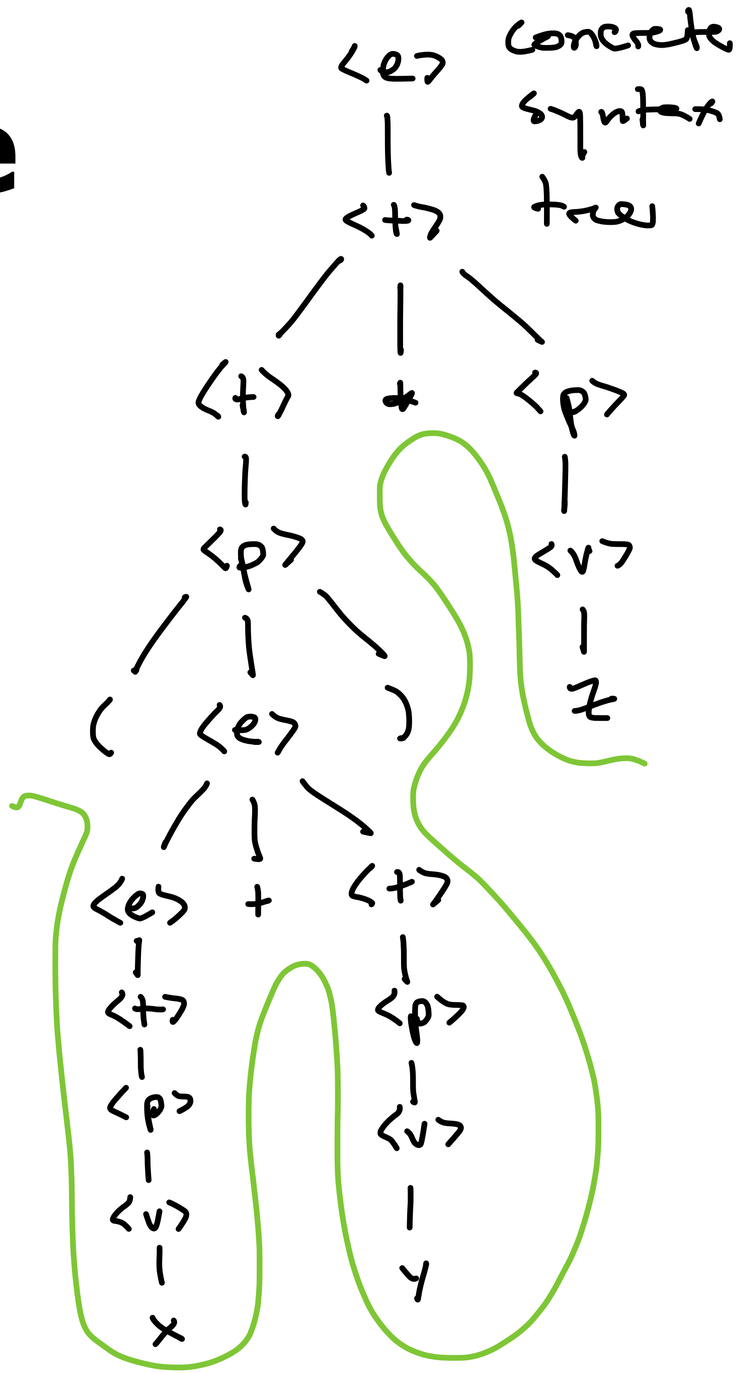
We further factor out the part of the rule for parentheses. Note that any expression can appear in the parentheses

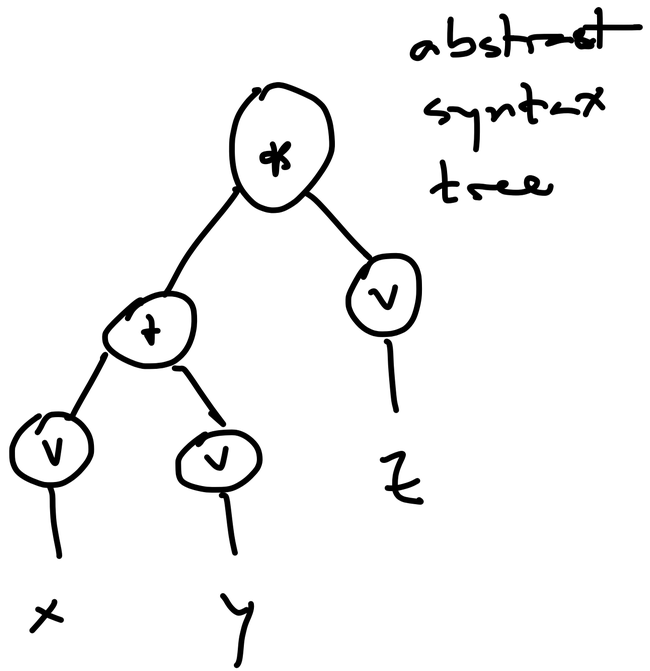
(This is a circular, or mutually recursive, production rule)

Example

```
<expr> ::= <expr> + <term>
         | <term>
<term> ::= <term> * <var>
         | <pars>
<pars> ::= <var> | ( <expr> )
<var>  ::= x | y | z
```

(x + y) * z





Summary

When we specify a PL (e.g., in the projects) you will be given a **BNF grammar**

You will need to know how to translate this into a parser

So you will need *practice reading BNF specifications*

To avoid ambiguity, we make choices beforehand about the **fixity, associativity** and **precedence**